

RESEARCH ARTICLES | 27 MAY 2026

Assembly theory: formalizing assembly spaces, discovering patterns and bounds

Wawrzyniec Bieniawski; Piotr Masierak; Andrzej Towski; Szymon Łukaszyk  ; Szymon Tworz[+ Author & article information](#)*R Soc Open Sci.* (2026) 13 (5): 260082 .<https://doi.org/10.1098/rsos.260082> **Article history** 

Related content

Review history: <https://www.webofscience.com/api/gateway/wos/peer-review/10.1098/rsos.260082> 

Abstract

Assembly theory defines structural complexity as the minimum number of steps required to construct an object in an assembly space. We formalize the assembly space as an acyclic digraph of strings. Key results include analytical bounds on the minimum and maximum assembly indices (ASIs) as functions of the string length and alphabet size, as well as relations between the ASI, assembly depth, depth index, Shannon entropy and expected waiting times for strings drawn from uniform distributions. We identify patterns in minimum- and maximum-ASI strings and provide construction methods for the latter. While computing ASI is NP-complete, we develop efficient implementations that enable ASI computation of long strings. We establish a counterintuitive, inverse relationship between a string ASI and its expected waiting time. Ordered decimal representations of low ASI bitstrings of even length N naturally cluster on diagonals and oblique lines of the squares with sides equal to $2^{N/2}$. Comparison with dictionary-based (LZW) and grammar-based (Re-Pair) compression schemes shows that ASI provides superior compression by exploiting global combinatorial patterns. These findings advance the use of complexity measures in computational biology, graph theory and data compression.

Keywords: assembly theory, information theory, complexity measures, mathematical physics, assembly index, graph theory, compression algorithms, information entropy, computational biology

Subjects: mathematical physics



1. Introduction

Assembly theory (AT) quantifies the minimum number of constructive steps required to generate an object, providing a principled measure of its structural complexity. It is built around two fundamental concepts: an assembly pool (a predefined set of basic items and a set of hitherto assembled subitems) and an assembly step (joining a pair of items taken from the assembly pool in a predefined order and cloning [1] such an assembled subitem to the assembly pool if it is not there), which together define the assembly index (ASI) as the minimal number of steps necessary to construct the object. Since its original formulation in 2017 [2], AT has been generalized to a broad range of theoretical and practical settings. A considerable body of work on AT [3–13] showcased the scale invariance of AT: its implementations, whether based on graphs, building blocks, DNA nucleobases, bounded structures or more abstract units assembled in various spaces, share the same fundamental characteristics.

Intrinsic purpose-driven mechanics implementing global combinatorics (heuristics) of AT provides a genuinely different perspective from classical notions of algorithmic or statistical complexity, despite the fact that AT appears to share many methodological features with classical dictionary-based [14–16] and grammar-based [17–32] data compression schemes. This resemblance stems from the fact that both AT and compression algorithms seek to capture redundancy, identify reusable substructures and derive concise generative descriptions of data. However, instead of quantifying randomness or compressibility, AT focuses on causal construction pathways and structural regularities encoded in minimal assembly pathways (i.e. in the ASI). This perspective naturally raises the question of how such regularities can be efficiently detected, represented and exploited by formal computational methods.

In this study, we focus on the general string version of AT. The paper is organized as follows. Section 2 introduces definitions and basic theorems used in the paper. Section 3 concerns the minimum complexity strings of AT, showing certain relations between the minimum ASI, assembly depth (ASD), depth index (DPI) and the Shannon entropy. Section 4 concerns the maximum complexity strings of AT, provides certain bounds of the maximum ASI as a function of a string length and the number of basic symbols, as well as certain methods of constructing high ASI strings. Section 5 concerns results common to the minimum and maximum complexity strings of AT, including a correlation between the expected waiting time (EWT) for a string of outcomes obtainable from a uniform categorical distribution and the ASI of this string. Section 6 discusses the supremacy of ASI compression over all polynomial-time compression algorithms. Finally, §7 summarizes the findings of this



study. In general, the proofs of theorems and lemmas, as well as certain methods and tables, are given in the electronic supplementary material.

2. General framework

Consider a string $C_k^{(N,b)}$ (we often write it simply as C_k) of length N made of b distinct basic symbols (unit-length strings) c . We shall label the symbols c using decimal digits $\{0, 1, \dots, b-1\}$ and letters if $b > 10$, where $b_{-k} := b - k$. We can assume that all strings C_k belong to a set C and are vertices of a graph, where certain strings are connected via edges e from a set E .

Definition 2.1. (Assembly Space). An assembly space $\Omega = (C, E)$ is an acyclic digraph of strings $C = \{C_k\}$, $k \in \mathbb{N}$, where all $b \in \mathbb{N}$ unit length strings (basic symbol(s)) are source vertices, and the remaining strings are 2-in-regular assembly step vertices, and E is a set of edges.

Definition 2.2. (Assembly Step). An assembly step s is the formation of a new string C_z in an assembly space Ω from two, not necessarily different, strings C_x, C_y by concatenating $C_z = C_x \circ C_y = \text{strcat}(C_x, C_y)$ which forms the edges $e = (C_x, C_z)$ and $e' = (C_y, C_z)$.

Since all the Ω vertices are strings, any edge $e = (C_x, C_z)$ is unambiguously defined. For example, the edge $e = ([010], [0101])$ unambiguously resolves to $e = ([010], [010] \circ [1])$ yielding $e' = ([1], [0101]) = ([1], [010] \circ [1])$ and $C_y = [1]$.

It is furthermore convenient to separate the set of strings into a set

$B := C \setminus \{C_k^{(N,b)} \in C : N \neq 1\}$ of source vertices and a set

$S := C \setminus \{C_k^{(N,b)} \in C : N \neq 1\}$ of 2-in-regular assembly steps vertices.

Definitions 2.1 and 2.2 are consistent: all vertices are unique (in any standard graph, all vertices should be unique), and all are strings. Since an assembly step always consists of joining two parts only [2], this can be thought of as the left and right fragments of the newly formed string, and those strings that can be the result of the concatenation of two shorter strings are assembly step 2-in-regular vertices. The uniqueness of each vertex is a sufficient criterion to establish the admissibility of an assembly step and to introduce the notion of an assembly pool: vertices (strings) present in the assembly space cannot be *assembled again* as new vertices of Ω , as they would not be unique.

Definition 2.3. (Assembly Plan). An assembly plan Ω_{C_s} of a string C_s is the assembly space 2.1 containing the vertex C_s and all the vertices leading to the string C_s .

There can be more than one assembly plan for a target string, reflecting different assembly pathways leading to it. However, the ASI $a^{(N,b)}(C_s)$ of a string $C_s^{(N,b)}$ is the minimum cardinality $|S(\Omega_{C_s})|$ of the set of the assembly step vertices $S(\Omega_{C_s})$ of all assembly plans Ω_{C_s} of the string C_s .

Strings in an assembly space 2.1 can form 2-grams, 3-grams, 4-grams, ..., n -grams, in general, wherein the following lemma [11] holds.

Lemma 2.1. *For all b , a 4-gram is the shortest string that allows for more than one ASI.*

$N = 2$ provides b^2 available 2-grams with unit ASI. $N = 3$ provides b^3 available 3-grams with ASI equal to two. Only $N = 4$ provides b^4 4-grams that include b^2 4-grams with ASI equal to two, that is b 4-grams $C_{k,\min}^{(4,b)} = [**]**$ and $b(b-1)$ 4-grams $C_{l,\min}^{(4,b)} = [**]**$, while the ASI of the remaining $b^4 - b^2$ 4-grams is three. For example, to assemble the 4-gram $C_{k,\min}^{(4,4)} = [0202]$, we need to assemble the 2-gram $[02]$ and reuse it, while there is nothing available to reuse, in the case of the 4-gram $C_{l,\min}^{(4,4)} = [0123]$.

Definition 2.4. (Clear/Mixed String). We call a string (an n -gram) containing only one symbol a clear string (a clear n -gram). We call a string (an n -gram) that is not clear a mixed string (a mixed n -gram).

Theorem 2.1. *Each k_r copies of an n_r -gram $C_r^{(n_r,b)}$ contained in a string $C_m^{(N,b)}$ decrease its ASI at least by $k_r(n_r - 1) - a^{(n_r,b)}(C_r)$. That is*

$$a^{(N,b)}(C_m) \leq N - 1 - \sum_{r=1}^R \left[k_r(n_r - 1) - a^{(n_r,b)}(C_r) \right], \quad (2.1)$$

where R is the total number of repeated n_r -grams.

For example, due to the presence of three copies of a 5-gram $[01001]$, each with $a^{(5,6)}([01001]) = 3$, in a string

$$C_k^{(24,6)} = [12|01001|21|01001|235|01001|52], \quad (2.2)$$

its ASI amounts to $a^{(24,6)}(C_k) = 24 - 1 - (3 \cdot (5 - 1) - 3) = 14$. The relation (2.1) provides the upper bound on ASI as it does not describe a situation in which an n -gram for $n > 2$ is assembled based on a $(n - 1)$ -gram also present in the string $C_m^{(N,b)}$, outside of this n -gram. For example, the ASI $a^{(14,9)}([56|101|78|101|43|01]) = 10$, while $14 - 1 - (2(3 - 1) - 2) = 11$.

Consider the strings C_k and C_l , where C_l is C_k in reverse order. The string C_k can have a distribution of n_r -grams leading to its ASI decrease (2.1). But the string C_l will

have the same distribution of n_r -grams in reverse order. Consequently, the strings C_k and C_l have the same ASI.

Definition 2.5. (AT String Equivalence). We consider two strings C_k and C_l to be AT-equivalent if they have the same ASI because C_l is C_k in reverse order and/or if C_l and C_k differ at most in symbol permutations.

If it is not otherwise clear from the context, in this study, we consider equivalent strings (e.g. $\{[001222], [111200], \dots\}$) as one string.

Another measure of a string's complexity is the ASD [33].

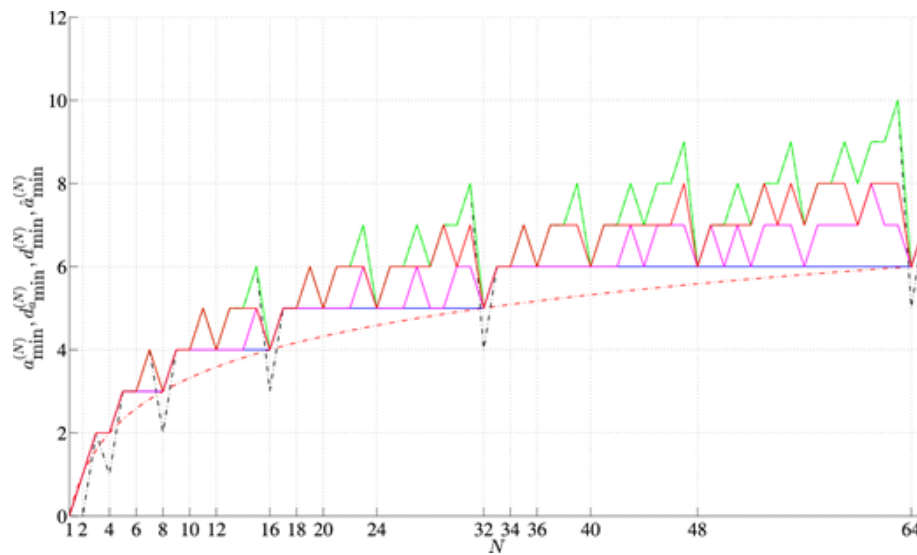
Definition 2.6. (ASD). The ASD

$$d_s^{(N_k, b)}(C_k) := \max \left(d^{(N_l, b)}(C_l), d^{(N_m, b)}(C_m) \right) + 1, \quad (2.3)$$

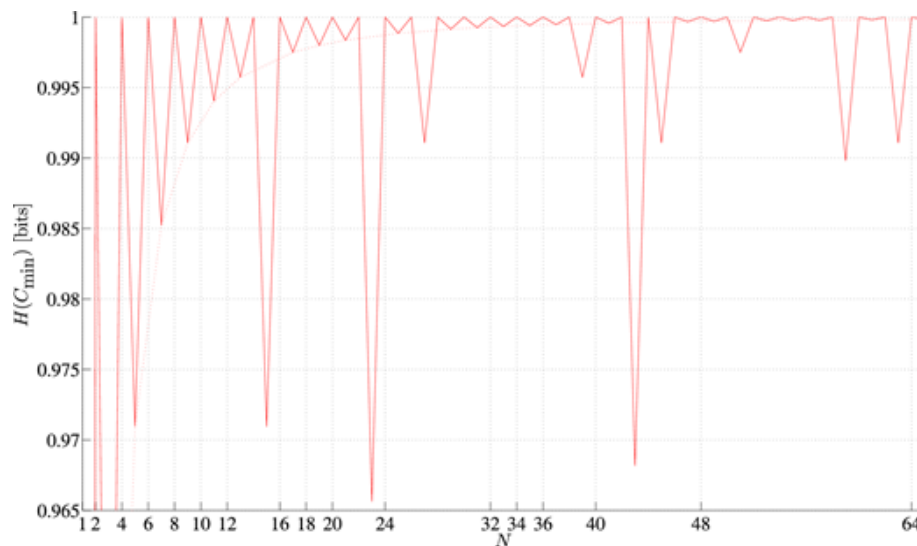
where $d_0^{(l, b)}(c) := 0$, and $d^{(N_l, b)}(C_l)$ and $d^{(N_m, b)}(C_m)$ are the ASDs of two substrings C_l, C_m of the string C_k that were joined in step s . For $N > 3$, and if there are more assembly pathways with different depths w_j leading to a string, which happens if at least two independent assembly steps are possible, the minimum pathway depth is the ASD of this string. Hence, the ASD captures the notion of an independent assembly step.

The ASD naturally divides the lengths of strings into sections $2^{d-1} < N \leq 2^d$ (cf. [figures 1](#) and [2](#)).



Figure 1.

The minimum ASD ($\lceil \log_2(N) \rceil$, blue), the ASD of the minimum ASI string (magenta), the minimum ASI (OEIS A003313, red; $\log_2(N)$, red, dash-dot), DPI (OEIS A014701, green) and $\lfloor \log_2(N) \rfloor + N_1 - 2$ (black, dash-dot), where N_1 is the Hamming weight of the binary representation of N (black) for $1 < N \leq 65$.

Figure 2.

The Shannon entropy of the most balanced bitstrings having the minimum ASI for $1 < N \leq 65$.

Theorem 2.2. *If an assembly space Ω contains strings having the same (non-zero) ASD, they were assembled in independent assembly steps.*

In other words, if two strings C_l, C_m in Ω have the same ASD, their assembly pathways are unrelated to each other; by defining [equation \(2.3\)](#), neither of them could have been used in the assembly pathway of the other.

Theorem 2.3. *For all b , the maximum length N of any string that can be assembled with the ASD $d_s^{(N)}$ (2.3) satisfies*

Lemma 2.2. *For all b , the minimum ASD (minASD), as a function of a string length N , is given by*

where $\lceil x \rceil$ denotes the ceiling function.

We cannot consider the ASD apart from the ASI. For example, the ASD of a string $C_{\max}^{(7,2)} = [0001110]$ is $d_{a_{\max}}^{(7,2)} = \lceil \log_2(7) \rceil = 3$, even though this string can be assembled in six steps with three larger pathway depths $w_6 \in \{4, 5, 6\}$ as

Similarly, the ASD of a string $C_{\max}^{(8,2)} = [00011101]$ is $d_{\max}^{(8,2)} = \lceil \log_2(8) \rceil = 3$ as

However, the non-maximum and non-minimum ASI string $C_k^{(8,2)} = [01001011]$ has only two 2-grams that can be assembled in independent steps. Hence, its ASD cannot be decreased to $\lceil \log_2(8) \rceil = 3$

$$\begin{array}{ll}
 [01] & d_1 = 1, & [01] & w_1 = 1, \\
 [11] & d_2 = 1, & [010] & w_2 = 2, \\
 [010] & d_3 = 2, & [010010] & w_3 = 3, \\
 [010010] & d_4 = 3, & [0100101] & w_4 = 4, \\
 [01001011] & d_5 = 4, & [01001011] & w_5 = 5.
 \end{array} \tag{2.9}$$

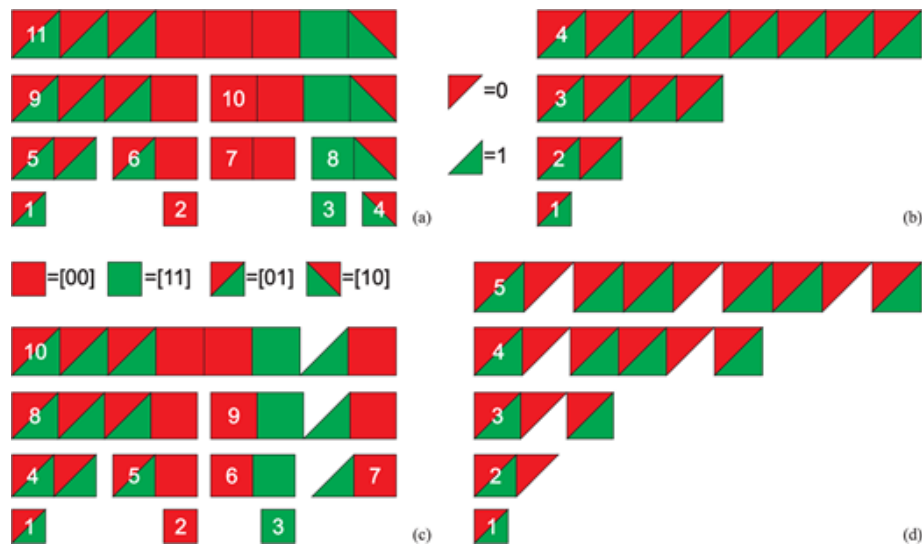
In general, the Ω that contains a 2^d -gram having the ASD d can also contain $\{2^{d-1} + 1, 2^{d-1} + 2, \dots, 2^d - 1\}$ -grams having the ASD d and based on the shorter n -grams of length $n < 2^{d-1} + 1$.

Theorem 2.4. For all b , the ASD of any maximum ASI string $C_{\max}^{(N,b)}$ corresponds to the minASD (2.5) of lemma 2.2, that is

$$d_{a_{\max}}^{(N,b)} = \lceil \log_2(N) \rceil. \tag{2.10}$$

For example, as shown in [figure 3c,d](#), the string $C_{\max}^{(15,2)} = [010101000011100]$ has the ASI $a_{\max}^{(15,2)} = 10$ and the ASD $d_{a_{\max}}^{(15,2)} = 4$, whereas the string $C_{\min}^{(15,2)} = [010010100101001]$ has smaller ASI $a_{\min}^{(15)} = 5$ but larger ASD $d_{a_{\min}}^{(15,2)} = 5$. On the other hand, the ASD of the maximum ASI string $C_{(N-5)}^{(16,2)} = [0101010000001110]$ and the minimum ASI string $C_{\min}^{(16,2)} = [0101010101010101]$, shown in [figure 3a,b](#), is the same.

Figure 3.



Assembly plans of bitstrings $C_{\max}^{(N,2)}$ (a,c) and $C_{\min}^{(N,2)}$ (b,d) for $N = 2^s = 16$ (a,b) and $N = 15 \neq 2^s$ (c,d), where the ASI is a number in a string (final string for (a,c)) and the ASD corresponds to the level. For $N = 2^s$, $d_{a_{\max}}^{(2^s,b)} = d_{a_{\min}}^{(2^s,b)} = s$. In general, for $N \neq 2^s$, the ASD $d_{a_{\max}}^{(N,b)} < d_{a_{\min}}^{(N,b)}$.

Here, we introduce the following definition, which—as we shall see—is also related to the independent assembly step.

Definition 2.7. (DPI). We call the number of steps $\hat{a}_{\min}^{(N)}$ to reach 1 starting from $N := N_{(0)}$ and assigning

$$N_{(s+1)} = \begin{cases} N_{(s)} - 1 & \text{if } N_{(s)} \text{ is odd,} \\ N_{(s)} - 2 & \text{if } N_{(s)} = 2^s + 2, s \in \mathbb{N}, \\ N_{(s)}/2 & \text{otherwise} \end{cases} \quad (2.11)$$

the DPI.

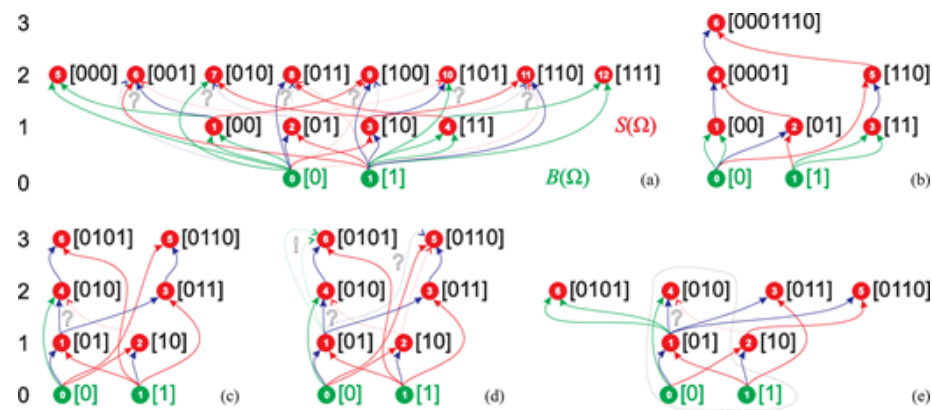
The relation (2.11) yields the same number of steps as the Chandah-sutra method (OEIS A014701) and, unlike the minimum ASI, is an analytical function of N . For example, $\hat{a}_{\min}^{(2^s)} = s$ and $\hat{a}_{\min}^{(2^s-1)} = 2(s-1)$.

Applying AT to evolutionary biology, we can assume that initially, a new string of length N is formed in an assembly space based on a basic symbol and a string of length $N-1$. Subsequently, this string assembly space evolves to reduce the cardinality $|S(\Omega_{C_s})|$ of the set of the assembly step vertices until it equals the ASI of this string, that is, until $|S(\Omega_{C_s})| = a^{(N,b)}(C_s)$. Hence, the assembly spaces evolve by reconfiguring the network of edges to decrease the ASD of newly assembled strings, possibly finding shorter pathways for these strings, and if only such a decrease would not result in ASI increase (length $N = 15$ shown in [figure 3d](#) is the shortest length, where $5 = d_{a_{\min}}^{(15)} > \lceil \log_2(15) \rceil = 4$).

The concepts of assembly space, assembly plan, ASI and ASD, as well as the evolution of assembly spaces, are illustrated in [figure 4](#). Although the alternative pathways shown in [figure 4a](#) demonstrate that the mixed 2-gram [01] is not required to construct all 3-grams in two steps, this 2-gram is necessary to achieve the ASI of longer strings (cf. a constrained AT* version shown in [table 1](#) discussed in §6).



Figure 4.



Assembly space, assembly plan, ASI and ASD. The assembly space of all eight binary 3-grams with all pathways (a). Blue edge provides the first string, red edge provides the second string in the assembly step and the order is irrelevant for two green edges or green edge provides the first or second string in dependence of the colour of the complementary edge. Dotted edges and question marks indicate alternative pathways. The optimal assembly plan of the bitstring $C_6^{(7,2)} = [0001110]$ showing that its ASI $\alpha^{(7,2)}(C_6) = 6$ (b). The evolution of assembly spaces of strings $[0101]$ and $[0110]$ (c–e). Strings $[0101]$ and $[0110]$ are initially assembled from 3-grams and basic symbols, increasing the ASD (c). New pathways increasing the number of independent assembly steps are found (d), and the edges of Ω are reconfigured, decreasing the number of assembly steps of the string $[0101]$ from three to two steps and the ASD of both 4-grams from three to two. The assembly space comprises five optimal assembly plans of the bitstrings $[0101]$, $[010]$ (two alternatives, one encircled), $[011]$ and $[0110]$ (e).

Table 1. Comparison of the ASI calculation, constrained ASI, Re-Pair and LZW compression plans.

step	ASI	ASI ^a	Re-Pair ^b	LZW ^c
1	[00]	[00]	[00] → A	[00] → C
2	[10]	[001]	[AA]	[000] → D
3	[0010]	[0010]	[AA1]	[10] → F
4	[000010]	[000010]	[AA1A]	[AC]
5	[0000100010]	[0000100010]	[AA1A0]	[AC A]
6			[AA1A01]	[AC AB]
7			[AA1 A010]	[AC ABD]
8				[AC ABDF]



	$\text{val}([0000100010]) = 34$
--	---------------------------------

step	ASI	ASI ^a	Re-Pair ^d	LZW ^e
1	[00]	[00]	[00] $\rightarrow$ A	[11] $\rightarrow$ C
2	[10]	[0000]	[11] $\rightarrow$ B	[00] $\rightarrow$ G
3	[110]	[00000]	[B1]	[BC]
4	[1110]	[100000]	[B10]	[BC A]
5	[111010]	[0100000]	[B101]	[BC AB]
6	[11101000]	[10100000]	[B101 A]	[BC ABA]
7	[1110100000]	[110100000]	[B101 AA]	[BC AB AG]
8		[1110100000]	[B101 AA0]	[BC AB AGG]
	$\text{val}([1110100000]) = 928$			

- a No 2-gram [10] allowed.
- b Dictionary $[00] \times 5 \rightarrow A$. Codeword [AA1A010].
- c Dictionary $[0] \rightarrow A$, $[1] \rightarrow B$, $[00] \rightarrow C$, $[000] \rightarrow D$, $[01] \rightarrow E$, $[10] \rightarrow F$ and $[0001] \rightarrow G$.
Codeword [ACABDF] containing only three dictionary elements (C, D, F) other than the basic symbols
 $|\text{Steps}| = |\text{Codeword}| - 1 + |\text{Dictionary}| = 6 - 1 + 3 = 8$.
- d Dictionary $[00] \times 4 \rightarrow A$, $[11] \times 2 \rightarrow B$. Codeword [B101AA0].
- e Dictionary
 $[0] \rightarrow A$, $[1] \rightarrow B$, $[11] \rightarrow C$, $[110] \rightarrow D$, $[01] \rightarrow E$, $[10] \rightarrow F$, $[00] \rightarrow G$ and $[000] \rightarrow H$.
Codeword [BCABAGG] containing only two dictionary elements (C, G) other than the basic symbols.
 $|\text{Steps}| = |\text{Codeword}| - 1 + |\text{Dictionary}| = 7 - 1 + 2 = 8$.

3. Minimum complexity strings of assembly theory

In this section, we consider factors relevant to the minimum string complexity in the context of AT. Theorem 3.1 was already stated in our previous study [11]; we restate it here for clarity.

Theorem 3.1.For all b , the minimum ASI (minASI) $a^{(N)}(C_{\text{min}})$ as a function of N corresponds to the shortest addition chain for N (OEIS A003313).

Any shortest addition chain for n starts with one, not zero, as zero is the neutral element of addition. For the same reason, two is considered the smallest prime, as one is the neutral element of multiplication. Hence, the fundamental theorem of arithmetic can be thought of as the shortest multiplication chain for N . It is conjectured that the largest prime factor must appear in some shortest addition chain for N [34].



Theorem 3.2. *The strings $C_{\min}^{(2^s, b)}$ can contain at most two distinct symbols if $b > 1$.*

Other minASI strings of length $N \neq 2^s$ can contain at most three distinct symbols if $b > 2$.

Proof. Minimum ASI strings of length $N = 2^s$ are formed by joining the newly assembled string to itself, where a clear or mixed 2-gram is assembled in the first step. Minimum ASI strings of other lengths admit a 2-gram and a 3-gram containing this 2-gram and an additional basic symbol.

To formally prove the first part, we can also use mathematical induction on the assembly step s . If $s = 1$, then the minASI strings $C_{\min}^{(2, b)}$ are 2-grams of the form $[c_1 c_2]$, where $c_1, c_2 \in B(\Omega)$. If $c_1 = c_2$, the string contains one distinct symbol, and if $c_1 \neq c_2$, the string contains two distinct symbols. In both cases, the number of distinct symbols does not exceed two. Now assume that for some $k \in \mathbb{N}$, all minASI strings $C_{\min}^{(2^k, b)}$ contain at most two distinct symbols. We must show that $C_{\min}^{(2^{k+1}, b)}$ also contains at most two distinct symbols. We construct $C_{\min}^{(2^{k+1}, b)}$ by joining two identical minASI strings $C_{\min}^{(2^k, b)}$,

$$C_{\min}^{(2^k, b)} \circ C_{\min}^{(2^k, b)} = C_{\min}^{(2^{k+1}, b)}, \quad (3.1)$$

with each other. By the inductive hypothesis, each $C_{\min}^{(2^k, b)}$ contains at most two distinct symbols. Therefore, their concatenation also contains at most two distinct symbols. By induction, for all $s \in \mathbb{N}$, the minASI string $C_{\min}^{(2^s, b)}$ contains at most two distinct symbols.

We will now show that other minASI strings of length $N \neq 2^s$ can contain at most three distinct symbols if $b > 2$. We provide the construction of minASI strings with three symbols. In the first step $s = 1$, we assemble a 2-gram $[c_1 c_2]$, where $c_1, c_2 \in B(\Omega)$ and $c_1 \neq c_2$. Next, we join the existing 2-gram $[c_1 c_2]$ with a new symbol $c_3 \in B(\Omega)$, where $c_3 \notin \{c_1, c_2\}$. This forms a 3-gram $[c_1 c_2 c_3]$, introducing a third distinct symbol and further increasing the ASI by 1. We continue assembling by joining the longest string formed so far with itself or with previously formed strings, maintaining the minimal ASI increase.

Assume *a contrario* that there exists a minASI string $C_{\min}^{(N, b)}$ of length $N \neq 2^s$ that contains four or more distinct symbols. But, incorporating such a fourth symbol is equivalent to assembling a maxASI 4-gram, which contradicts the minimality of $C_{\min}^{(N, b)}$ (only a 2-gram must be assembled from basic symbols, and a 3-gram must be assembled from a basic symbol and a 2-gram). Thus, theorem 3.2 is proven. ■

By theorem 2.1, the maximum ASI decrease is provided by minASI 2^s -grams and amounts to $k(2^s - 1) - s$. On the other hand, 1-symbol n -grams are guaranteed to

have the minimum ASI for all n . Therefore, the maxASI string will tend to contain only one such long n -gram for each b . We have found, for example, that the sum of the lengths of an n_0 -gram of zeros and an n_1 -gram of ones should preferably be greater than or equal to $N/2$ of a maxASI string.

The assembly plans of the minASI strings of length $N \neq 2^s$ are not unique [35]. For example, a string $C_{\min}^{(5,b)} = [01010]$ can be assembled in three steps in four assembly plans with $S(\Omega) = \{[01], [010]\}$, $S(\Omega) = \{[01], [0101]\}$, $S(\Omega) = \{[10], [010]\}$ or $S(\Omega) = \{[10], [1010]\}$.

Strings with non-minASI can contain all symbols. For example, the string [36]

$$C_k = [01234012340123401234] \quad (3.2)$$

has ASI $a^{(20,5)}(C_k) = 6 = a_{\min}^{(20)} + 1$ and contains all five basic symbols $B(\Omega) := \{0, 1, 2, 3, 4\}$.

The minASD as a function of the length of a string $d_{\min}^{(N)}$ (2.5), the ASD of a minASI string $d_{a_{\min}}^{(N)}$ (which we call here the *minASI ASD*), the minASI as a function of the length of a string $a_{\min}^{(N)}$ (OEIS A003313) and DPI $\hat{a}_{\min}^{(N)}$ (OEIS A014701) define four distinct sets illustrated in [figure 1](#), wherein $d_{\min}^{(N)} \leq d_{a_{\min}}^{(N)} \leq a_{\min}^{(N)} \leq \hat{a}_{\min}^{(N)}$. We observed certain salient regularities among them.

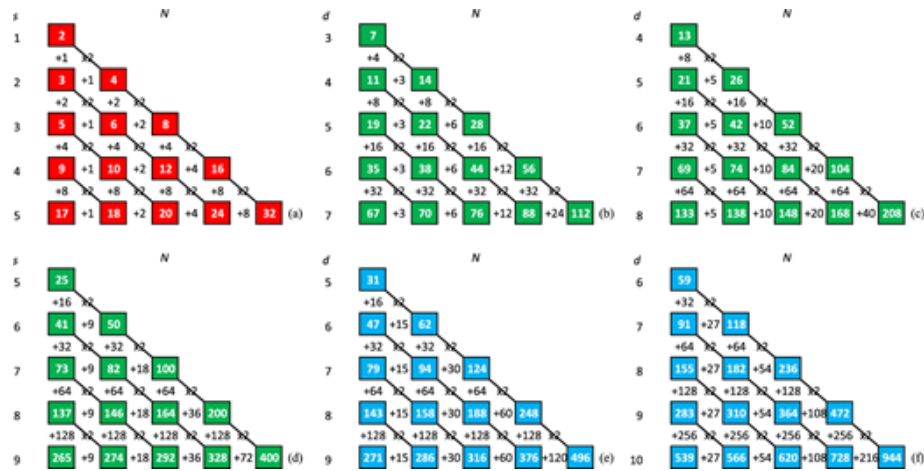
Theorem 3.3. *If a minASI string has length $N := 2^s$, $s \in \mathbb{N}_0$, then the minimum ASD, minASI ASD, minASI and DPI are equal to s .*

Theorem 3.3 can be generalized as follows.

Theorem 3.4. *The minASD, minASI ASD, minASI and DPI of a minASI string are equal to $s \in \mathbb{N}$ if and only if $\hat{N}_1 := 2^{s-1} + 2^l$, $l = 0, 1, \dots, s-1$, $s \geq 1$ or, in other words (cf. [figure 5a](#))*



Figure 5.



Lengths of all strings having the property of $d_{\min}^{(\tilde{N}_1)} = a_{\min}^{(\tilde{N}_1)} = d_{a_{\min}}^{(\tilde{N}_1)} = \hat{a}_{\min}^{(\tilde{N}_1)} = s$

(a). Lengths $\tilde{N}_3, \tilde{N}_5, \tilde{N}_9$ of certain strings having the property of

$a_{\min}^{(\tilde{N}_*)} = d_{\min}^{(\tilde{N}_*)} + 1$ (b–d). Lengths $\tilde{N}_{15}, \tilde{N}_{27}$ of certain strings having the property

of $a_{\min}^{(\tilde{N}_*)} = d_{\min}^{(\tilde{N}_*)} + 2$ (e,f).

$$\hat{N}_1 := 2^{s-1} + 2^l, \quad l = 0, 1, \dots, s-1, \quad s \geq 1 \quad \Leftrightarrow \quad d_{\min}^{(\hat{N}_1)} = d_{a_{\min}}^{(\hat{N}_1)} = a_{\min}^{(\hat{N}_1)} \quad (3.3)$$

The assembly plans of other minASI strings can contain independent assembly steps. The first such case occurs for $N = 7$, where, for example, the $S(\Omega)$

$$\begin{aligned} [01] \quad d_1 &= 1, \\ [001], [0101] \quad d_2 &= d_3 = 2, \\ [0010101] \quad d_4 &= 3 \end{aligned} \quad (3.4)$$

results in a string having $a_{\min}^{(7)} = 4$ and $d_{a_{\min}}^{(7)} = \lceil \log_2(7) \rceil = 3$, since both $[001]$ and $[0101]$ were assembled from the 2-gram $[01]$ in two independent assembly steps at the same depth $d_2 = d_3 = 2$, which is congruent with theorem 2.2.

Lemma 3.1. *The minASI strings $[11]$ (strings (15)) of lengths*

$$\tilde{N}_3 := 2^{d-1} + 3 \cdot 2^l, \quad l = 0, 1, \dots, d-3, \quad d \geq 3 \quad \Leftrightarrow \quad a_{\min}^{(\tilde{N}_3)} = d + 1 = \lceil \quad (3.5)$$

have only one independent assembly step in their assembly plans; excluding this step, they are assembled by joining the longest string assembled so far to itself. Therefore, their ASI is one greater than the minASD (2.5) (cf. figure 5b).

Lemma 3.2. *The minASI strings of lengths*

$$\tilde{N}_5 := 2^{d-1} + 5 \cdot 2^l, \quad l = 0, 1, \dots, d-4, \quad d \geq 4 \quad \Leftrightarrow \quad a_{\min}^{(\tilde{N}_5)} = d+1 = \left\lceil \right\rceil \quad (3.6)$$

have only one independent assembly step in their assembly plans; excluding this step, they are assembled by joining the longest string assembled so far to itself. Therefore, their ASI is one greater than the minASD (2.5) (cf. [figure 5c](#)).

Lemma 3.3. *The minASI strings of lengths*

$$\tilde{N}_9 := 2^{d-1} + 9 \cdot 2^l, \quad l = 0, 1, \dots, d-5, \quad d \geq 5 \quad \Leftrightarrow \quad a_{\min}^{(\tilde{N}_9)} = d+1 = \left\lceil \right\rceil \quad (3.7)$$

have only one independent assembly step in their assembly plans; excluding this step, they are assembled by joining the longest string assembled so far to itself. Therefore, their ASI is one greater than the minASD (2.5) (cf. [figure 5d](#)).

Lemmas 3.1–3.3 allow for the following generalization.

Theorem 3.5. *The minASI strings of lengths*

$$\tilde{N}_{2^{n+1}} := 2^{d-1} + (2^{k-4} + 1)2^l, \quad k \geq 5, \quad d \geq k-2, \quad l = 0, 1, \dots, d-(k-2) \quad (3.8)$$

$$\Leftrightarrow \quad a_{\min}^{(\tilde{N}_{2^{n+1}})} = d+1 = \left\lceil \log_2(\tilde{N}_{2^{n+1}}) \right\rceil + 1$$

have only one independent assembly step in their assembly plans; excluding this step, they are assembled by joining the longest string assembled so far to itself. Therefore, their ASI is one greater than the minASD (2.5).

The lengths of the strings (3.8) are listed in rows in [table 2](#), starting after the length of the substring assembled in an independent assembly step marked green. Hence, the first row contains the lengths of strings of lemma 3.1 shown on the diagonal of [figure 5b](#), and so on.

Table 2. Certain lengths of minASI strings, which are defined by the ASI and the minASI ASD for $2 \leq s \leq 7$.

	minASI										
s	1	2	3	4	5	6	7	8	9	...	$\tilde{N}_{2^{n+1}}$
2	2	4	<u>3</u>	7	14	28	56	112	224	...	$\tilde{N}_3$
					15	30	60	120	240	480	$\tilde{N}_{3,a}$
						23	46	92	184	368	$\tilde{N}_{3,b}$

3	2	4	8	<u>3</u>	11	22	44	88	176	...	$\tilde{N}_3$
						27	54	108	216	432	$\tilde{N}_{3,a}$
							43	86	172	344	$\tilde{N}_{3,b}$
	2	4	8	<u>5</u>	13	26	52	104	208	...	$\tilde{N}_5$
						45	90	180	360	...	$\tilde{N}_{5,b}$
4	2	4	8	16	<u>3</u>	19	38	76	152	...	$\tilde{N}_3$
							51	102	204	408	$\tilde{N}_{3,a}$
								83	166	332	$\tilde{N}_{3,b}$
	2	4	8	16	<u>5</u>	21	42	84	168	...	$\tilde{N}_5$
							85	170	340	...	$\tilde{N}_{5,b}$
	2	4	8	16	<u>9</u>	25	50	100	200	...	$\tilde{N}_9$
5	2	4	8	16	32	<u>3</u>	35	70	140	...	$\tilde{N}_3$
								99	198	396	$\tilde{N}_{3,a}$
									163	326	$\tilde{N}_{3,b}$
	2	4	8	16	32	<u>5</u>	37	74	148	...	$\tilde{N}_5$
								165	330	...	$\tilde{N}_{5,b}$
	2	4	8	16	32	<u>9</u>	41	82	164	...	$\tilde{N}_9$
	2	4	8	16	32	<u>17</u>	49	98	196	...	$\tilde{N}_{17}$
6	2	4	8	16	32	64	<u>3</u>	67	134	...	$\tilde{N}_3$
									195	390	$\tilde{N}_{3,a}$
										323	$\tilde{N}_{3,b}$
	2	4	8	16	32	64	<u>5</u>	69	138	276	$\tilde{N}_5$
									325	650	$\tilde{N}_{5,b}$
	2	4	8	16	32	64	<u>9</u>	73	146	...	$\tilde{N}_9$
	2	4	8	16	32	64	<u>17</u>	81	162	...	$\tilde{N}_{17}$
	2	4	8	16	32	64	<u>33</u>	97	194	...	$\tilde{N}_{33}$



7	2	4	8	16	32	64	128	<u>3</u>	131	...	$\tilde{N}_3$
										387	$\tilde{N}_{3,a}$
	2	4	8	16	32	64	128	<u>5</u>	133	266	$\tilde{N}_5$
										645	$\tilde{N}_{5,b}$
	2	4	8	16	32	64	128	<u>9</u>	137	...	$\tilde{N}_9$
	2	4	8	16	32	64	128	<u>17</u>	145	...	$\tilde{N}_{17}$
	2	4	8	16	32	64	128	<u>33</u>	161	...	$\tilde{N}_{33}$
	2	4	8	16	32	64	128	<u>65</u>	193	...	$\tilde{N}_{65}$

Theorem 3.6. The minASI strings [\[11\]](#) of lengths

$$\begin{aligned} \tilde{N}_7 &:= 2^{d-1} + 7 \cdot 2^{d-4} \in \{15, 30, 60, \dots\}, \quad d \geq 4 \quad \Leftrightarrow \\ a_{\min}^{(\tilde{N}_7)} &= d_{a_{\min}}^{(\tilde{N}_7)} = d_{\min}^{(\tilde{N}_7)} + 1 = \hat{a}_{\min}^{(\tilde{N}_1)} - 1 = \left\lceil \log_2(\tilde{N}_7) \right\rceil + 1 \end{aligned} \quad (3.9)$$

are assembled by joining the longest string assembled so far with itself. Their ASI and ASD are the same: one greater than the minASD [\(2.5\)](#) and one smaller than the DPI.

Theorem 3.6 seems to allow for the following generalization, which we have validated numerically based on the shortest addition chain sequence for $N \leq 10^5$.

Conjecture 3.1. For d, l and $\tilde{N}_{2^n+1}$ defined by the relation [\(3.8\)](#), the following holds:

$$\begin{aligned} \tilde{N}_{2^n+1,a} &:= \tilde{N}_{2^n+1} + 2^d = 3 \cdot 2^{d-1} + (2^{k-4} + 1)2^l \wedge k = 5, \\ \Leftrightarrow \quad a_{\min}^{(\tilde{N}_{2^n+1,a})} &= d + 2 = \left\lceil \log_2(\tilde{N}_{2^n+1,a}) \right\rceil + 1, \end{aligned} \quad (3.10a)$$

$$\begin{aligned} \tilde{N}_{2^n+1,b} &:= \tilde{N}_{2^n+1} + 2^{d+1} = 5 \cdot 2^{d-1} + (2^{k-4} + 1)2^l \wedge k \in \{5, 6\}, \\ \Leftrightarrow \quad a_{\min}^{(\tilde{N}_{2^n+1,b})} &= d + 3 = \left\lceil \log_2(\tilde{N}_{2^n+1,b}) \right\rceil + 1. \end{aligned} \quad (3.10b)$$

The lengths of the strings [\(3.10a\)](#) and [\(3.10b\)](#) are listed in rows in [table 2](#).

Furthermore, we have numerically validated the following conjecture.

Conjecture 3.2. The minASI strings of lengths

$$\tilde{N}_{15} := 2^{d-1} + 15 \cdot 2^l, \quad l = 0, 1, \dots, d-5, \quad d \geq 5, \quad (3.11a)$$

$$\tilde{N}_{27} := 2^{d-1} + 27 \cdot 2^l, \quad l = 0, 1, \dots, d-6, \quad d \geq 6, \quad (3.11b)$$

$$\tilde{N}_{50.9} := 50 \cdot 2^{d-6} + 9 \cdot 2^l, \quad l = 0, 1, \dots, d-6, \quad d \geq 6, \quad (3.11c)$$

— the assembly plan of a maxASI string will tend to maximize the number of strings assembled in independent assembly steps in the Ω , taking into account the saturation of the Ω as it cannot contain more than b^n distinct n -grams, and hence to minimize the possible ASD.

We note that, in general, the difference between the DPI and minASI is larger than 1.

Lemma 3.4. *The minASI bitstrings assembled along the pathway given by the DPI 2.7 and beginning with $C_{\min}^{(2)} = [\star\star]$ are balanced bitstrings if N is even or nearly balanced bitstrings ($N_0 = N_1 + 1$) if N is odd.*

In other words, the DPI removes the *imbalance propagation*. For example, an imbalanced pathway $\{2, 4, 5, 10, 20\}$ ($N_1 = 8$) becomes a balanced pathway $\{2, 4, 8, 10, 20\}$ ($N_1 = 10 = N/2$) and $2^s + 2 \rightarrow 2^{s-1} + 1 \rightarrow 2^{s-1} \Leftrightarrow 2^s + 2 \rightarrow 2^s \rightarrow 2^{s-1}$. We have also examined the Shannon entropy

$$H(C_{\min}^{(N)}) = -p_0 \log_2(p_0) - p_1 \log_2(p_1) \quad (3.15)$$

of the most balanced minASI bitstrings, where $p_0 = N_0/N$ and $p_1 = N_1/N$ are fractions of the respective symbols $\{0, 1\}$ within the string (N_1 is the Hamming weight). Although by theorem 3.1, the minASI as a function of the length of a string does not depend on b , we have chosen the most balanced bitstrings, as they maximize the Shannon entropy (3.15). This choice is also supported by physics [37–39]. For the same reason, we preferred the pathway $\{2, 3, 5, 10, 15\}$ (cf. figure 3d) over $\{2, 3, 6, 12, 15\}$, for example, as the string assembled using the former pathway is more balanced ($N_1 = 6$) than the one assembled using the latter one ($N_1 = 5$). Similarly, we preferred the pathway that provided a more balanced string over the one that provided independent assembly steps. $N = 14$ is the first exception. $C_{\min}^{(14)}$ assembled in five steps along the pathway $\{2, (3, 4), 7, 14\}$ with the independent assembly steps 3 and 4 has the Hamming weight $N_1 = 6$ as compared to $C_{\min}^{(14)}$ assembled in five steps along the pathway $\{2, 4, 8, 12, 14\}$ with no independent assembly steps and the Hamming weight $N_1 = N/2 = 7$. The resulting pathways of the minASI strings maximizing the number of independent assembly steps or the binary Shannon entropy (3.15) are listed in table 3 for $N \leq 65$. As shown in figure 2, the Shannon entropy (3.15) of the most balanced minASI bitstrings rapidly converges to one with exceptions for lengths $N \in \{15, 23, 27, 39, 43, 45, 51, 59, 63, \dots\}$ substantially corresponding to lengths at which DPI is larger than the minASI (cf. figure 1), which highlights the interdependence between the minASI and DPI.

Table 3. Pathways leading to minASI strings (maximizing the number of independent assembly steps—MIA, maximizing the binary Shannon entropy—MBL) for $2 \leq N \leq 65$.

N	$d_{\min}^{(N)}$	$d_{a\min}^{(N)}$	$a_{\min}^{(N)}$	$\hat{a}_{\min}^{(N)}$	MIA pathway	MBL pathway (N_1)
2	1	1	1	1	{2} (1)	
3	2	2	2	2	{2, 3} (1)	
4	2	2	2	2	{2, 4} (2)	
5	3	3	3	3	{2, 4, 5} (2)	
6	3	3	3	3	{2, 4, 6} (3)	
7	3	3	4	4	{2, (3, 4), 7}	{2, 4, 6, 7} (3)
8	3	3	3	3	{2, 4, 8} (4)	
9	4	4	4	4	{2, 4, 8, 9} (4)	
10	4	4	4	4	{2, 4, 8, 10} (5)	
11	4	4	5	5	{2, (3, 4), 7, 11}	{2, 4, 8, 10, 11} (5)
12	4	4	4	4	{2, 3, 6, 12}	{2, 4, 8, 12} (6)
13	4	4	5	5	{2, 4, (5, 8), 13}	{2, 4, 8, 12, 13} (6)
14	4	4	5	5	{2, (3, 4), 7, 14}	{2, 4, 8, 12, 14} (7)
15	4	5	5	6	{2, 3, 5, 10, 15} (6)	
16	4	4	4	4	{2, 4, 8, 16} (8)	
17	5	5	5	5	{2, 4, 8, 16, 17} (8)	
18	5	5	5	5	{2, 4, 8, 16, 18} (9)	
19	5	5	6	6	{2, (3, 4), 8, 11, 19}	{2, 4, 8, 10, 18, 19} (9)
20	5	5	5	5	{2, 3, 5, 10, 20}	{2, 4, 8, 16, 20} (10)
21	5	5	6	6	{2, 4, (5, 8), 16, 21}	{2, 4, 8, 16, 20, 21} (10)
22	5	5	6	6	{2, (3, 4), 7, 11, 22}	{2, 4, 8, 16, 20, 22} (11)
23	5	6	6	7	{2, 3, 5, 10, 20, 23} (9)	
24	5	5	5	5	{2, 4, 8, 12, 24} (12)	
25	5	5	6	6	{2, 4, 8, (9, 16), 25}	{2, 4, 8, 16, 24, 25} (12)
26	5	5	6	6	{2, 4, (5, 8), 13, 26}	{2, 4, 8, 16, 24, 26} (13)
27	5	6	6	7	{2, 3, 6, 12, 24, 27}	{2, 4, 5, 9, 18, 27} (12)



28	5	5	6	6	$\{2, (3, 4), 7, 14, 28\}$	$\{2, 4, 8, 16, 24, 28\}$ (14)
29	5	6	7	7	$\{2, 4, 8, (9, 10), 20, 29\}$	$\{2, 4, 8, 16, 24, 28, 29\}$ (14)
30	5	6	6	7	$\{2, 3, 5, 10, 15, 30\}$	$\{2, 4, 6, 10, 20, 30\}$ (15)
31	5	6	7	8	$\{2, 4, (5, 8), 13, 26, 31\}$	$\{2, 4, 8, 10, 20, 30, 31\}$ (15)
32	5	5	5	5	$\{2, 4, 8, 16, 32\}$ (16)	
33	6	6	6	6	$\{2, 4, 8, 16, 32, 33\}$ (16)	
34	6	6	6	6	$\{2, 4, 8, 16, 32, 34\}$ (17)	
35	6	6	7	7	$\{2, (3, 4), 7, 14, 28, 35\}$	$\{2, 4, 8, 16, 32, 34, 35\}$ (17)
36	6	6	6	6	$\{2, 4, 8, 16, 32, 36\}$ (18)	
37	6	6	7	7	$\{2, 4, (5, 8), 16, 32, 37\}$	$\{2, 4, 8, 16, 32, 36, 37\}$ (18)
38	6	6	7	7	$\{2, (3, 4), 8, 11, 19, 38\}$	$\{2, 4, 8, 16, 32, 36, 38\}$ (19)
39	6	6	7	8	$\{2, 4, (5, 8), 13, 26, 39\}$	$\{2, 4, (5, 8), 13, 26, 39\}$ (18)
40	6	6	6	6	$\{2, 4, 8, 16, 32, 40\}$	$\{2, 4, 8, 16, 32, 40\}$ (20)
41	6	6	7	7	$\{2, 4, 8, (9, 16), 25, 41\}$	$\{2, 4, 8, 16, 32, 40, 41\}$ (20)
42	6	6	7	7	$\{2, (3, 4), 7, 14, 28, 42\}$	$\{2, 4, 8, 16, 32, 40, 42\}$ (21)
43	6	7	7	8	$\{2, 3, 5, 10, 20, 40, 43\}$ (17)	
44	6	6	7	7	$\{2, (3, 4), 7, 11, 22, 44\}$	$\{2, 4, 8, 16, 32, 40, 44\}$ (22)
45	6	7	7	8	$\{2, 3, 5, 10, 20, 40, 45\}$	$\{2, 4, 5, 9, 18, 27, 45\}$ (20)
46	6	7	7	8	$\{2, 3, 5, 10, 20, 23, 46\}$	$\{2, 4, 6, 10, 20, 40, 46\}$ (23)
47	6	7	8	9	$\{2, (3, 4), 7, 11, 22, 44, 47\}$	$\{2, 4, 6, 10, 20, 40, 46, 47\}$ (23)
48	6	6	6	6	$\{2, 4, 8, 12, 24, 48\}$ (24)	
49	6	7	7	7	$\{2, 4, 8, 12, 24, 48, 49\}$ (24)	
50	6	6	7	7	$\{2, 4, 8, (9, 16), 25, 50\}$	$\{2, 4, 8, 16, 32, 40, 48, 50\}$ (25)
51	6	7	7	8	$\{2, 4, 8, 16, 17, 34, 51\}$ (24)	
52	6	6	7	7	$\{2, 4, (5, 8), 13, 26, 52\}$	$\{2, 4, 8, 16, 32, 40, 48, 52\}$ (26)
53	6	7	8	8	$\{2, 4, (5, 8), 16, 32, 48, 53\}$	$\{2, 4, 8, 16, 32, 40, 48, 52, 53\}$ (26)
54	6	7	7	8	$\{2, 3, 6, 12, 24, 27, 54\}$	$\{2, 4, 6, 12, 24, 48, 54\}$ (27)
55	6	7	8	9	$\{2, (3, 4), 7, 11, 22, 44, 55\}$	$\{2, 4, 8, 16, 18, 36, 54, 55\}$ (27)
56	6	6	7	7	$\{2, (3, 4), 7, 14, 28, 56\}$	$\{2, 4, 8, 16, 32, 48, 56\}$ (28)

57	6	7	8	8	$\{2, (3, 4), 7, 14, 28, 56, 57\}$	$\{2, 4, 8, 16, 32, 48, 56, 57\}$ (28)
58	6	7	8	8	$\{2, (3, 4), 7, 14, 28, 29, 58\}$	$\{2, 4, 8, 16, 32, 48, 56, 58\}$ (29)
59	6	7	8	9	$\{2, (3, 4), 7, 14, 28, 56, 59\}$	$\{2, 4, 5, 9, 18, 27, 54, 59\}$ (26)
60	6	7	7	8	$\{2, 4, 8, 12, 24, 48, 60\}$	$\{2, 4, 6, 10, 20, 30, 60\}$ (30)
61	6	8	8	9	$\{2, 4, 8, 12, 24, 48, 60, 61\}$	$\{2, 4, 8, 16, 20, 40, 60, 61\}$ (30)
62	6	7	8	9	$\{2, (3, 4), 7, 14, 28, 31, 62\}$	$\{2, 4, 8, 16, 20, 40, 60, 62\}$ (31)
63	6	7	8	10	$\{2, (3, 4), 7, 14, 21, 42, 63\}$	$\{2, 4, 5, 9, 18, 27, 45, 63\}$ (28)
64	6	6	6	6	$\{2, 4, 8, 16, 32, 64\}$ (32)	
65	7	7	7	7	$\{2, 4, 8, 16, 32, 64, 65\}$ (32)	

We demonstrated in this section that minimum complexity AT strings cannot be fully derived by any rule-based system. The same is true for the maxASI strings, as we shall show in a subsequent section.

4. Maximum assembly index strings of assembly theory

A seven-bit string is the longest string that can have the maxASI $a_{\max}^{(7,2)} = N - 1 = 6$. It contains two clear 3-grams

$$C_{(N-1)}^{(7,2)} = [0001110], \quad (4.1)$$

and its length cannot be increased without a repetition of a 2-gram, which keeps the ASI at the same level $a_{\max}^{(8,2)} = 8 - 2 = 6$. This led us to seek a general method to construct the longest possible string having the maxASI $a_{\max}^{(N,b)}(C_{(N-1)}) = N - 1$, as a function of the radix b . We denote the length of this string by $N_{(N-1)}$, and we call this string a $C_{(N-1)}$ string. We developed two methods of constructing a string $C_{(N-1)}$ of the form

$$C_{(N-1)} = [0001112 \dots b_{-2}b_{-1}b_{-1}b_{-1} \dots 0]. \quad (4.2)$$

In both methods, we start with an initial balanced string of length $3b$ containing b clear 3-grams ordered as

$$[0001112 \dots b_{-2}b_{-1}b_{-1}b_{-1}]. \quad (4.3)$$

The 2-grams that can be inserted into the initial string (4.3) can be arranged in a $b \times b$ matrix

$$\begin{bmatrix} \cancel{00} & \cancel{01} & 02 & \dots & 0b_{-1} \\ 10 & \cancel{11} & \cancel{12} & \dots & 1b_{-1} \\ 20 & 21 & \cancel{22} & \dots & 2b_{-1} \\ \dots & \dots & \dots & \dots & \dots \\ b_{-2}0 & b_{-2}1 & b_{-2}2 & \dots & \cancel{b_{-2}b_{-1}} \\ b_{-1}0 & b_{-1}1 & b_{-1}2 & \dots & \cancel{b_{-1}b_{-1}} \end{bmatrix}, \quad (4.4)$$

where the crossed-out entries on a diagonal cannot be reused, as they would form repetitions in this string. Due to the order of 3-grams in the string (4.3), we can also cross out the entries in the first superdiagonal of the matrix (4.4). By construction, the starting string (4.3) has length $3b$ and contains only $b + (b - 1) = 2b - 1$ 2-grams $[l]$ and $[m(m + 1)]$, where $0 \leq l \leq b_{-1}$ and $0 \leq m \leq b_{-2}$.

In the first step of the first method, we begin by forming a string containing 2-grams on the first subdiagonal of the matrix (4.4), starting with 10:

$$[102132 \dots b_{-2} b_{-3} b_{-1} b_{-2}], \quad (4.5)$$

and we append it to the string (4.3). With this step, we also eliminate the 2-grams on the second superdiagonal starting with the 2-gram 02, as well as the 2-gram $b_{-1}1$. In the second step, we form a string containing 2-grams on the third superdiagonal beginning with the 2-gram 03:

$$[0314 \dots b_{-5} b_{-2} b_{-4} b_{-1}], \quad (4.6)$$

and append it to the string formed so far. With this step, we also remove the 2-gram $b_{-2}0$ and the middle part of the second subdiagonal containing $\{31, 42, \dots, b_{-2}b_{-4}\}$. And so on. Finally, we append 0 if b is even. This process is illustrated in [figure 6](#) and generates the following $C_{(N-1)}^{(N,b)}$ strings:

$$\begin{aligned} C_{(N-1)}^{(3,1)} &= [000], \\ C_{(N-1)}^{(7,2)} &= [000111|0], \\ C_{(N-1)}^{(13,3)} &= [000111222|10|20], \\ C_{(N-1)}^{(21,4)} &= [000111222333|102132|03|0], \\ C_{(N-1)}^{(31,5)} &= [000111222333444|10213243|0314|20|40], \\ C_{(N-1)}^{(43,6)} &= [000111222333444555|1021324354|031425|0415|2053|0], \\ C_{(N-1)}^{(57,7)} &= [000111222333444555666|102132435465|03142536|041526|206 \\ &\dots \end{aligned}$$



Figure 1 displays a series of 16 small grids (0-15) illustrating the evolution of a 10x10 cellular automaton over time. Each grid has a top header with numbers 0-9 and a left header with numbers 0-9. The grids are color-coded: red for 0, green for 1, blue for 2, yellow for 3, orange for 4, light green for 5, light blue for 6, light orange for 7, light green for 8, and light blue for 9. The grids show a progression from a simple initial state (0) to a complex, chaotic state (15).

Subsequently, we considered other strings with the maxASI. A $C_{(N-1)}$ string contains all 2-grams. Hence, inserting any basic symbol into any position inevitably leads to a repetition of a 2-gram. Without loss of generality, we append it at the start of the $C_{(N-1)}$ string, obtaining a string $[*000111 \dots]$. Another symbol can be introduced to this string without an additional 2-gram repetition, for example, as another prefix $[\star * 000111 \dots]$ leading to the repetition of the 2-gram $\star\star$ or $\star 0$ but not both of them (here we allow $\star = *$). Hence, both the length and the ASI of this string increase by one. Finally, 0 can be appended at the start of this string without an additional 2-gram repetition, provided that $\star = 1$ and $*$ = 0. For $b = 2$, there are only four¹ bitstrings containing only two copies of a 2-gram having the ASI $a_{\max}^{(10,2)} = 8$:

$$[01 | 0000 | 1110], \quad [1 | 0000 | 11101], \quad [101 | 0000 | 111], \quad [1 | 0000 |$$

where the leftmost one is the only unbalanced string within this set ($N_0 \neq N_1$). Each 10-bit string (4.9) has one clear 4-gram surrounded by the same symbol ($[\dots 1|0000|1 \dots]$) and one clear 3-gram ($[\dots 111 \dots]$), which can be concatenated ($[\dots 1|0000|111 \dots]$) in eight bits or separated ($[\dots 1|0000|1 \dots 111 \dots]$) in nine bits. The first three strings (4.9) correspond to the former case, while the last one corresponds to the latter. Each of the strings (4.9) contains three pairs of mutually exclusive 2-grams, so that only one pair can be reused to decrease the maximum $N - 1$ ASI by one.

We developed two methods of constructing generalized $C_{(N-b)}$ strings (4.9) for $b > 2$. They have $b - 1$ clear 4-grams ($c_l c_l c_l c_l$), one clear 3-gram and $(b - 1) \times 3$ of mutually exclusive 2-grams $\{c_k c_l, c_l c_l, c_l c_m\}$ overlapped in patterns $[\dots c_k | c_l c_l c_l c_l | c_k \dots]$ surrounding a clear 4-gram, so that only one pair from each pattern can be reused in 3^{b-1} ways to reduce the maxASI to $N - b$. The first method is based on the balanced $C_{(N-2)}$ string (4.9) $[101|0000|111]$ and generates the most balanced $C_{(N-b)}$ strings. It starts with a string of length $6b - 2$ containing all $b - 1$ clear 4-grams in the form

$$[b_{-1}01 | 0000 | 12 | 1111 | 2 \dots b_{-1} | b_{-2}b_{-2}b_{-2} | b_{-1}b_{-1}b_{-1}] \quad (4.10)$$

to which the missing $b^2 - 2b$ symbols are appended using the patterns shown in [figure 7](#), yielding the $C_{(N-b)}$ strings of the form



Figure 7.



Two-gram matrices for $1 \leq b \leq 9$ that illustrate the generation of the most balanced $C_{(N-b)}$ strings (4.10). Coloured 2-grams are appended to the initial string of clear 4-grams in the order indicated by colours.

$$C_{(N-1)}^{(3,1)} = [000],$$

$$C_{(N-2)}^{(10,2)} = [101|0000|111],$$

$$C_{(N-3)}^{(19,3)} = [201|0000|12|1111|222|102],$$

$$C_{(N-4)}^{(30,4)} = [301|0000|12|1111|23|2222|333|103|202|13],$$

$$C_{(N-5)}^{(43,5)} = [401|0000|12|1111|23|2222|34|3333|444|104|3031\ 3202|14$$

$$C_{(N-6)}^{(58,6)} = [501|0000|12|1111|23|2222|34|3333|45|4444|555|105|4041]$$

• • •

$$C_{(N-b)}^{(b^2+4b-2,b)} = [b-1 \ 01|0000|1 \dots b-1|b-2 \ b-2 \ b-2|b-1 \ b-1 \ b-$$

• • •

We discuss the second method, generating non-balanced $C_{(N-b)}$ strings, in the electronic supplementary material.

The mutually exclusive 2-grams of any $C_{(N-b)}$ string comprise $b - 1$ adjoining clear 2-grams on the diagonal of the 2-grams matrix (4.4) and $2(b - 1)$ 2-grams on its sub- and superdiagonals, as shown in [figure 7](#). Thus, because each clear 4-gram is surrounded by the same symbol, only one pair from each pattern can be reused. In particular, $C_{(N-b)}$ strings can have a form (4.11) (see also the electronic supplementary material) with all clear 4-grams separated by 2-grams and forming 7-gram patterns in the form $[\dots c_l c_k c_l c_l c_l c_l c_k \dots]$. However, this is not necessary. For example, the string $C_{(N-3)} = [1|0000|102|1111|21|222|01]$ does not follow this rule. The $C_{(N-b)}$ strings are not the maximum length maxASI strings that have the property of maxASI incremented by one with every three symbols. For example, the string (4.11) for $b = 3$ can be suffixed with 202, yielding a string with $a^{(22,3)} = 18$, which still is not the maxASI string with this property for $b = 3$ (cf. [table 4](#)).

Table 4. Exemplary maxASI strings for $b = 3$ and $13 \leq N \leq 54$.

N	$C_{\max}^{(N,3)}$	$a_{\max}^{(N,3)}$
13	[0002220111210]	12

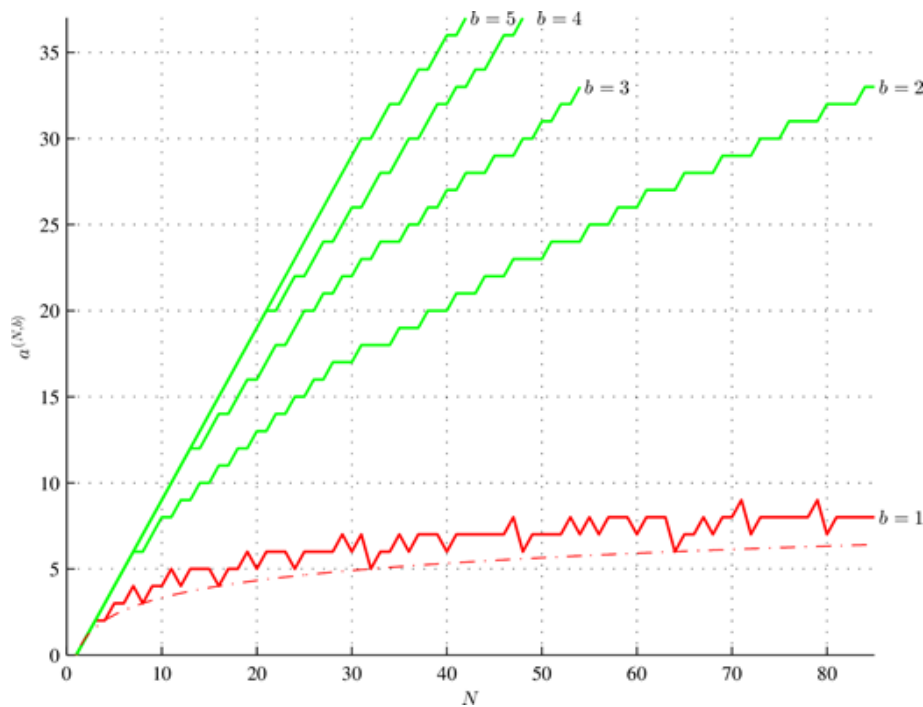
14	[00022201112101]	12
15	[000222011121012]	13
16	[0002220111210120]	14
17	[20011121002201021]	14
18	[222111210100001202]	15
19	[02211101001222000021]	16
20	[022111010012220000211]	16
21	[0221110100122200002112]	17
22	[0221110100122200002021]	18
23	[022111010012220000211201]	18
24	[101212200022210211100120]	19
25	[0222212112002010001111021]	20
26	[02222121120020100011110210]	20
27	[012221211200201000111102202]	21
28	[212001112011022201000201210]	21
29	[01222121120020100011110220210]	22
30	[012221211200201000111102202102]	22
31	[0122212112002010001111002202102]	23
32	[01222121120020100011110220210200]	23
33	[0122212112002010000001111022021020]	24
34	[0122212112002010001111022021020010]	24
35	[01222121120020100011110220210200000]	24
36	[012221211200201000111102202102000202]	25
37	[012221211200201000111102202011121021]	25
38	[01222121120020100011110220210202000010]	26
39	[012221211200201000111102202102020002210]	26
40	[0122212112002010001111022021020010110000]	27
41	[01222121120020100011110220210200101100002]	27
42	[012221211200201000111102202102001011000002]	28

43	[0122212112002010001111022021020010110000222]	28
44	[01222121120020100011110220210200101100000211]	28
45	[012221211200201000111102202102001010211100002]	29
46	[2111020110222211012201112212121010020000001202]	29
47	[21110201102222110122011122121210100120210021200]	29
48	[211102011022221101220111221212101002000000120210]	30
49	[2111020110222211012201112212121010020000001202102]	30
50	[21110201102222110122011122121210100200000012021021]	31
51	[211102011022221101220111221212101002000000120211210]	31
52	[2111020110222211012201112212121010020000001202112012]	32
53	[21110201102222110122011122121210100200000012021120102]	32
54	[211102011022221101220111221212101002000000120211200220]	33

5. Results common to the minimum and maximum complexity strings

The minASI and maxASI bounds are illustrated in [figure 8](#) for $b = \{2, 3, 4\}$. For each b , maxASI initially equals $N - 1$ up to the first threshold [\(4.8\)](#). Then it flattens and decreases by one every 3 symbols: once for $b = 2$ and $2b$ times for $b > 2$, as we conjecture based on the results for $b = 3$.



Figure 8.

The minimum ASI (red; $\log_2(N)$, red, dash-dot) and the maximum ASI (green) for $b \in \{2, 3, 4, 5\}$.

The ASI of a string is also related to the EWT of this string. The EWT is the expected number of trials until a specific string appears in a sequence of independent and identically distributed random variables taking values in a finite set of $b > 1$ equiprobable outcomes, called a uniform categorical distribution. The general formula for the EWT of a string, which can be determined using Markov chains, is

[41]

$$\mathbb{E} [C_l^{(N,b)}] = \sum_{k=1}^N b^k \cdot \mathbf{I}_k, \quad (5.1)$$

where $\mathbf{I}_k$ is an indicator function that is 1 if the prefix of length k is also the suffix of the string $C_l^{(N,b)}$ and 0 otherwise. Thus, the EWT is bounded from below and from above as

$$b^N \leq \mathbb{E} [C_l^{(N,b)}] \leq \frac{b}{b-1} (b^N - 1). \quad (5.2)$$

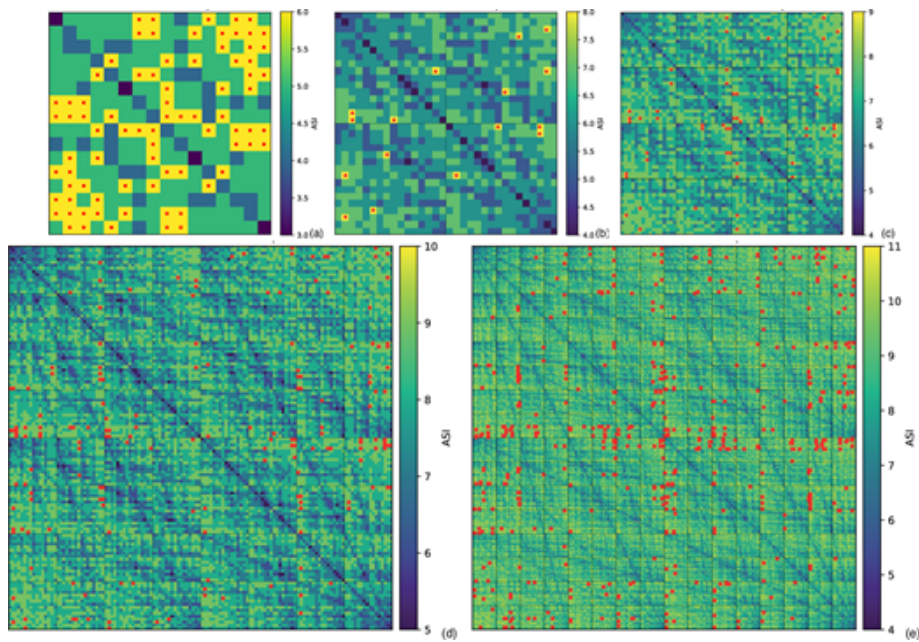
The upper EWT bound (5.2) is attained by clear strings, that is, also by the minASI strings, in which case $\mathbf{I}_k = 1, \forall k$ and the formula (5.1) simplifies to a geometric series $\mathbb{E} [C_l^{(N,b)}] = \sum_{k=1}^N b^k$, which resolves to the upper EWT bound (5.2). The minASI strings containing two or three symbols do not satisfy this relation and hence have shorter EWTs. For example, the minASI alternating strings have the EWT given

by $E[C_l^{(N,b)}] = \sum_{m=1}^{N/2} b^{2m}$ if N is even and $E[C_l^{(N,b)}] = \sum_{m=1}^{\lceil N/2 \rceil} b^{2m-1}$ if N is odd, that is, $E[C_k^{(N,b)}] \in \{2, 4, 10, 20, 42, 84, \dots\}$ given by $\{\times 2, \times 2 + 2, \times 2, \dots\}$ (OEIS A026644). The non-alternating minASI bitstrings can have the EWT even closer to the lower bound (5.2). For example, $E[010010100101001] = 33828$ (cf. figure 3d) lies firmly at the lower end of the spectrum, being positioned just 3.24% of the way through the total range. The lower EWT bound (5.2) is attained by the strings having no prefixes matching suffixes; in which case, for $k = N$, the whole string matches itself. The maxASI strings feature the EWT values equal to or close to this bound.

Applying AT to evolutionary biology, the inverse proportionality between the EWT of a string and its ASI is anticipated, but also feels counterintuitive, as common sense would suggest that strings of low complexity should be *provided earlier* than more complex ones.

We found a correlation between the maxASI and the decimal representation of a string, providing an intuitive geometric interpretation of the complexity landscape. For example, figure 9 shows the ASI of all bitstrings of even lengths $N \in \{8, 10, 12, 14, 16\}$ distributed in squares of sides equal to $2^{N/2}$, starting from the top-left corner ($\text{val}_2[00 \dots] = 0$) and ending at the bottom-right corner ($\text{val}_2[11 \dots] = 2^N - 1$). The decimal value of each pixel is given by $r2^{N/2} + c$, where $0 \leq r, c \leq 2^{N/2} - 1$ are the row and column indices. The low ASI strings naturally occur on diagonals and oblique lines because of the internal symmetries they exhibit. For example, minASI strings naturally occur on the main diagonal, as $r(2^{N/2} + 1)$ for $r = c$ corresponds to a bitstring containing a prefix of length $N/2$ equal to its suffix. Thus, the EWT (5.1) of a main diagonal bitstring C_d is bounded from below by $E[C_d^{(N,2)}] \geq 2^N + 2^{N/2}$, and its ASI, taking into account the relation (2.1), is bounded by



Figure 9.

The ASI of all bitstrings of lengths $N = 8$ (a), $N = 10$ (b), $N = 12$ (c), $N = 14$ (d) and $N = 16$ (e), as a function of their decimal representations, in squares with sides equal to $\sqrt{2^N}$. Red dots indicate maxASI strings.

$$a_{\min}^{(N,2)} \leq a^{(N,2)}(C_d) \leq N - 1 - 2(N/2 - 1) + a_{\max}^{(N/2,2)} = a_{\max}^{(N/2,2)} + 1. \quad (5.3)$$

For example, for $N = 8$, the ASI of all main diagonal strings C_d satisfy $3 \leq a^{(8,2)}(C_d) \leq 4$, and they include the four minASI strings $\{[00 \dots], [01 \dots]\} \times 2$ corresponding to equidistributed decimals 0, 85, 170 and 255. For $N = 10$, the depleted 16 maxASI strings correspond to {94, 122, 270, 286, 376, 450, 482, 488, 535, 541, 573, 647, 737, 753, 901, 929} decimal values. For $N = 12$, the 10 minASI strings $\{[00 \dots], [001 \dots], [010 \dots], [01 \dots], [011 \dots]\} \times 2$ are equidistributed on the main square diagonal every 585 and every 1365. Finally, for $N = 16$, the small ASI strings are aligned on diagonals and skewed lines with the four minASI strings again equidistributed on the main square diagonal, while the 506 maxASI strings tend to attain row-like and column-like arrangements.

6. Assembly index compression versus polynomial-time compression algorithms

Calculating the ASI of a string can be contrasted with compression algorithms that run in polynomial time, such as basic dictionary-based (LZ-type) schemes [42–44] or grammar-based algorithms, such as Re-Pair. Indeed, Turing's universality and algorithmic complexity show that any computable transformation, including the ASI, can be simulated symbolically with no loss of information [42]. This naturally raised

doubts regarding the trivial computability of the ASI and its potential equivalence to popular lossless compression algorithms.

Yet these doubts proved moot. As the length of a string increases, computing the ASI becomes a hyper-exponentially hard, exhaustive process over all possible combinations and repeated substrings due to the enormous combinatorial space of possible assembly plans and the need to globally evaluate which fragments are most efficient to create early versus later; such a pathway minimization cannot be resolved entirely through local heuristics; there is no local strategy of finding optimal substrings in AT. This problem is widely known in AT literature (see [13,45–47]). It was conjectured [11] that the ASI problem is NP-complete. A certificate-dependent (assembly-step) proof of this conjecture within predefined assembly plans [12,46] was found [47] by providing a Karp reduction from the vertex cover problem. A complete, self-contained proof of NP-completeness of the ASI problem for both decision and optimization variants was found [48] by establishing a correspondence between assembly plans and straight-line grammars. Since, for each ASI, there exists a naive algorithm (exhaustive search of all the assembly plans), the ASI is computable for all finite objects [7]. Thus, one can say that determining ASI is computable and algorithmically trivial. But it is also computationally non-trivial because the assembly tree grows exponentially with N . One can see this non-triviality by noting that integrating machine-learning-driven methods does not improve ASI computation, even for moderately sized molecules [49]. Interestingly, human intuition can often identify structural patterns or regularities [11], enabling rapid non-algorithmic qualitative estimation of the ASI that remains inaccessible to sequential algorithms.

The computational non-triviality of the ASI raised a second question: whether a worse and slower algorithm [42] (such as NP-complete ASI exhaustive search) merits consideration if it provides the same explanatory power as known algorithms. However, the answer to this question is again negative, as we illustrate with the following simple examples. Without loss of generality, we compare:

- i. the ASI calculation;
- ii. a constrained ASI calculation version that prohibits the assembly of a 2-gram [10] in an assembly plan;
- iii. the grammar-based Re-Pair method, which counts all 2-grams in a string, selects the most frequent 2-gram that occurs at least twice, replaces all its occurrences with a new symbol and repeats these steps until no 2-gram occurs more than once, while forming a codeword containing the respective symbols; and

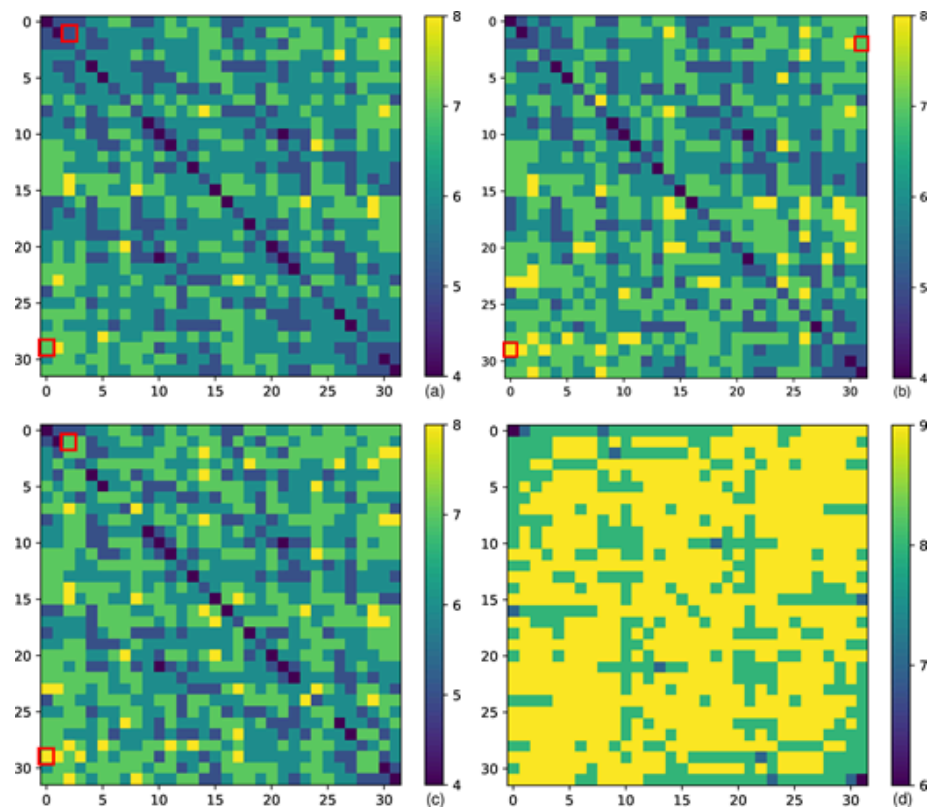
iv. the dictionary-based LZW method, which searches for a first prefix of a string that does not exist in the dictionary (initially containing single symbols), adds this first prefix as a new dictionary entry and adds this first prefix without its last symbol to the codeword, and repeats this search in a string, starting from the last symbol of this prefix until the string is exhausted,

for all 2024 bitstrings of length $N = 10$.

To compare the ASI calculation with another compression algorithm, we can only compare the ASI with the number of steps achievable in the assembly plan generated by this algorithm. The distributions of steps illustrated in [figure 10](#) (and elaborated in [table 1](#) for two exemplary strings) show that ASI compression is superior to that provided by the Re-Pair or LZW algorithms. Furthermore, [figure 10](#) shows that if additional constraints are imposed on the ASI calculation, then it may yield the same, or even worse, results as compared to known compression algorithms, given the large number of the latter.



Figure 10.



The ASI (a), constrained ASI (b), Re-Pair steps (c) and LZW steps (d) of all bitstrings of length $N = 10$, as a function of their decimal representations ($\{0, 1, \dots, 1023\}$), in squares with sides equal to 32 pixels. Visible lack of perfect correlation between these compressions: the number of Re-Pair steps is higher than ASI for 100 strings (e.g. $7 = \text{Re-Pair}(34) > \text{ASI}(34) = 5$ and $8 = \text{Re-Pair}(928) > \text{ASI}(928) = 7$, respectively, for $[0000100010]$ and $[1110100000]$). Also visible is lack of horizontal symmetry for constrained ASI (e.g. $7 = a_{\text{cons}}^{(10,2)}([0001011111]) \neq a_{\text{cons}}^{(10,2)}([1110100000]) = 8$ if 2-gram $[10]$ is excluded from the assembly plans).

We encourage the reader to conduct an independent comparison of the ASI (e.g. for the maxASI strings provided in [tables 4–7](#)) with the results produced by well-known compression algorithms.

Table 5. Exemplary maxASI bitstrings for $10 \leq N \leq 39$.

N	$C_{\text{max}}^{(N,2)}$	$a_{\text{max}}^{(N,2)}$	N	$C_{\text{max}}^{(N,2)}$	$a_{\text{max}}^{(N,2)}$
10	[1010000111]	8	25	[000000010101011001111110]	15
11	[00010111100]	8	26	[0100110000011111101010110]	16
12	[101010000111]	9	27	[00000001111110101011001000]	16
13	[1000001110101]	9	28	[011010101111110000000110010]	17



14	[10011000010111]	10	29	[011000000010101001110111110]	17
15	[000001010111110]	10	30	[10010000000011001010110111111]	17
16	[1001100001010111]	11	31	[01010100100000001111110101100]	18
17	[0000001010111110]	11	32	[010011000000001010110111111001]	18
18	[100110100001010111]	12	33	[100000000010011111101110101010]	18
19	[011110110000010100]	12	34	[1000000000100111111011101010101]	18
20	[1001101000010101111]	13	35	[1010100000001011001001111110001101]	19
21	[00000001010100111110]	13	36	[10101000000010110010011111100011101]	19
22	[001011111101001100000]	14	37	[101101010100000001001000111111001101]	19
23	[00000001010101100111110]	14	38	[1011101010100000001001000111111001101]	20
24	[01100111111010100000001]	15	39	[111001100100011010000001010101101111]	20

Table 6. Exemplary maxASI bitstrings for $40 \leq N \leq 85$.

N	$C_{\max}^{(N,2)}$	$a_{\max}^{(N)}$
40	[00110010111111010100000011011000101101]	20
41	[0000011111001110101001011011011000100110]	21
42	[00110111110101010110000000111100100100101]	21
43	[011110011110110010100000011100011000101101]	21
44	[11101010101011111100100100011000000010110110]	22
45	[111010101010111111001001000110000000111011010]	22
46	[011110011111010001110000000110010010101001101]	22
47	[01111001111101000111000000011001001010110110]	23
48	[011110011111010001110000000110010010101101100]	23
49	[0111100111110100011100000001100100101011011000]	23
50	[101001111110001000111101000000101101100001100110]	23
51	[10100111111000100011110100000010110110010011010101]	24
52	[101001111110001000111101000000101101100100110001101]	24
53	[101001111110001000111101000000101101100100110101010]	24
54	[1010011111100010001111010000001011011001001110111101]	24



55	[101001111110001000111101000000101101100100110001110101]	25
56	[10100111111100010001111010000001011011001001110111010101]	25
57	[1010011111110001000111101000000101101100100100110101010]	25
58	[10100111111100010001111010000001011011001001001101011000]	26
59	[101001111111000100011110100000010110110010010011010110000]	26
60	[101010111011110011111101001000011001000100010010110000000]	26
61	[101010111011110011111101001000011001000100011011010000000001]	27
62	[1010101110111100111111010010000110010001000100101100000001101]	27
63	[101010111011110011111101001000011001000100010010110110000000]	27
64	[1010101110111100111111010010000110010001000100101101100000001]	27
65	[10101011101111001111110100100001100100010001001011011000000001]	28
66	[101010111011110011111101001000011001000100010010110110000000011]	28
67	[101010101011001000111101101111100010000001100110011100110100100101]	28
68	[1010101010110010001111011011111000100000011001100111001101001010001]	28
69	[0110101010110010100100101110011000000011111101100100111011111010001]	29
70	[101010101011001000111101101111100010000001100110011100111000101001001]	29
71	[1010101010110010001111011011111000100000011001100111001110100100101001]	29
72	[10101010101100100011110110111110001000000110011001110010011101001100010]	29
73	[101010101011001000111101101111100010000001100110011100100111010010110001]	30
74	[1010101010110010001111011011111000100000011001100111001001110101100010100]	30
75	[10101010101100100011110110111110001000000110011001110011100010100100000001]	30
76	[011010101011001010010010111001100000001111110110010011100000100011000010111]	31
77	[0110101010110010100100101110011000000011111101100100111000001000110000101111]	31
78	[10101010101100100011110110111110001000000110011001110011100010100100000000001]	31
79	[100101110110101111110100110110010011101010110010100011001110111100000001011]	31
80	[1001011101101011111101001101100100111010101100101000110011101111000000010101]	32
81	[10010111011010111111010011011001001110101011001010001100111011110000000101010]	32
82	[10010111011010111111010011011001001110101011001010001100111011110000000101011]	32
83	[100101110110101111110100110110010011101010110010100011001110111110100000000111]	32

84	[100101110110101111111010011011001001110101011010010100011001111011110000000101011100]	33
85	[1001011101101011111110100110110010011101010110100101000110011110111100000001010111000]	33

Table 7. Exemplary maxASI strings for $b = 4$ and $21 \leq N \leq 49$.

N	$C_{\max}^{(N,4)}$	$a_{\max}^{(N,4)}$
21	[000111222333102132030]	20
22	[0001112223331021320302]	20
23	[00011122233310213203012]	21
24	[010000111222333102132030]	22
25	[0100001112223331021320302]	22
26	[01000011122233310213203023]	23
27	[010000111222333102132030221]	24
28	[0001102013331121301222230323]	24
29	[00011320133311121022232302030]	25
30	[301000012111123222233310320213]	26
31	[3010000121111232222333103202130]	26
32	[30100001211112322223331032021303]	27
33	[301000012111123222233310320200213]	28
34	[3010000121111232222333203102133130]	28
35	[30100001211112322223332031021331300]	29
36	[301000012111123222233320310213313110]	30
37	[3010000121111232222333203102133131101]	30
38	[30100001211112322223332031021331311011]	31
39	[301000012111123222233320310213313110221]	32
40	[3010000121111232222333203102133131102210]	32
41	[30100001211112322223332031021331311022101]	33
42	[301000012111123222233320310213313110322011]	33
43	[3010000121111232222333203102133131102210103]	34
44	[30100001211112322223332031021331311022101030]	34



45	[301000012111123222233320310213313110221201300]	35
46	[3010000121111232222333203102133131102212013002]	36
47	[30100001211112322223332031021331311022120130023]	36
48	[301000012111123222233320310213313110221201300230]	37
49	[3010000121111232222333203102133131102212013002303]	37

7. Discussion

The present study provides a general mathematical framework for the general AT string version and demonstrates how this formalization enables efficient computation of ASIs across various diverse structural sequences. Clarifying the theory’s conceptual structure provides suggestions for its computational implementation. This allowed us to formalize bounding and pruning processes (an approach also proposed in [13,45]) and to develop software capable of effectively computing the ASI of a string, significantly reducing the number of potential configurations to explore. This, in turn, allowed us to find the exemplary maxASI strings (cf. tables 4–7) and determine the maxASI bounds shown in figure 8, overcoming important limitations exhibited by all the other known implementations of the algorithms to compute the ASI of *big objects* [50].

Fortunately, the non-triviality of computing the ASI does not close the field but rather delineates a structured landscape for algorithmic exploration. The framework developed here is equally applicable to symbolic representations of various real-world patterns within the context of AT, such as detecting biosignatures [6], describing the evolution of music [51], and, as we conjecture, natural language processing (artificial intelligence-generated text detection, authorship style analysis, plagiarism detection), time series analysis, financial data changes (distinction between random and predictable behaviour), sensor data anomaly analysis, medical signal analysis, etc.

Furthermore, AT offers a unifying language for describing how information becomes structured. It can do so by showing how biological sequences are governed by foundational mathematical principles, for instance, for understanding the rules underlying the assembly of macromolecules such as proteins and DNA. Evolutionary biology explains the survival and prevalence of certain traits, but it does not address the mechanisms for generating novel phenotypic variants. We have shown, for example, that a DNA strand of length N containing four nucleobases cannot represent a minASI string without violating Chargaff’s rules and theorem 3.2.



Theorem 3.2 establishes that a minASI string can contain at most three distinct symbols (if $N \neq 2^s$, as otherwise it can contain at most two), while the first Chargaff’s rule (for double-stranded DNA) states that four nucleobases A, T, G and C should be pairwise balanced. On the other hand, biological systems tend to deviate from the equilibrium of maximum entropy [11,52] corresponding to balanced strings,

For $b = 1$, the information entropy (3.15) vanishes, as the bit ($b = 2$) is the smallest amount and the quantum of information. Yet AT explains the assembly of such minASI strings, extending beyond the minimal two-valued system necessary to convey any information. Traditional physics, while offering predictive power from past initial conditions to future states, lacks a functional perspective necessary to differentiate meaningful novelty from random fluctuations [53]. Our results clarify that AT is a formal system for describing the generative and causal structure of objects, positioning it as a theoretical extension of information theory, emphasizing not the description of information, but the effort required to construct it. Indeed, the better the compression, the more likely a model or theory is to capture a natural phenomenon, and the more accurately it may predict its behaviour [42].

Glossary

The following abbreviations are used in this paper:	
AT	assembly theory
N	length of a string
b	number of basic symbols $c \in \{0, 1, \dots, b - 1\}$
$C_s, C_s^{(N,b)} \in C$	a string (vertex)
$B := C \setminus \{C_k^{(N,b)} \in C : N \neq 1\}$	source vertices (base symbols)
$S := C \setminus \{C_k^{(N,b)} \in C : N = 1\}$	assembled strings (2-in-regular assembly step vertices)
$e = (C_x, C_z), e' = (C_y, C_z) \in E$	directed edges from strings C_x to C_z and C_y to C_z (possibly $C_x = C$)
$\Omega = (C, E)$	assembly space
ΩC_s	assembly plan of a string C_s
$\text{ASI}, a^{(N,b)}(C_s)$	assembly index of a string C_s (minASI—minimum, maxASI—maximum)
$\text{ASD}, d_s^{(N_k,b)}(C_s)$	assembly depth of a string C_s (minASD—minimum, minASI ASD—ASD)
DPI	depth index (OEIS A014701)
$\text{EWT}, E[C_s^{(N,b)}]$	expected waiting time of a string C_s



The following OEIS sequences are referenced in this paper:	
A003313	length of the shortest addition chain for n (minASI)
A014701	number of multiplications to compute n th power by the Chandah-sutr
A026644	number of moves to solve the Chinese rings puzzle
A048645	integers with one or two 1-bits in their binary expansion
A173786	triangle read by rows: $T(n, k) = 2^n + 2^k, 0 \leq k \leq n$

Ethics

This work did not require ethical approval from a human subject or animal welfare committee.

Data accessibility

Data and relevant code for this research work are stored in GitHub [\[54\]](#) and have been archived within the Zenodo repository [\[55\]](#).

Supplementary material is available online [\[56\]](#).

Declaration of AI use

We have not used AI-assisted technologies in creating this article.

Authors’ contributions

W.B.: conceptualization, data curation, formal analysis, investigation, methodology;
P.M.: conceptualization, formal analysis, investigation, methodology; A.T.: formal analysis, investigation, methodology, validation; S.L.: conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, project administration, resources, software, supervision, validation, visualization, writing—original draft, writing—review and editing; S.T.: conceptualization, investigation, software, visualization.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Conflict of interest declaration

We declare we have no competing interests.



Funding

No funding has been received for this article.

Supplement

Supplementary material is available online at

<https://doi.org/10.6084/m9.figshare.c.8490332>.

Acknowledgements

The authors thank Mariola Bala for her motivation, Rafał Bobrowski for assessing the time required to calculate a maxASI of a string as a function of its length, Robert Dobosz for providing valuable hints on earlier research, Rafał Winiarski for noting that the relation (2.1) is an inequality, an anonymous referee for pointing out the coincidence of the DPI with $\lceil \log_2(N) \rceil + N_1 - 2$, shown in [figure 1](#) and S.Ł.'s daughter Anna for her insight in finding the pattern of strings (equation (4.11)). We acknowledge Polish high-performance computing infrastructure PLGrid for awarding this project access to the LUMI supercomputer, owned by the EuroHPC Joint Undertaking, hosted by CSC (Finland) and the LUMI consortium through the pllatsl02 (19070) grant.

Notes

- ¹ Sixteen if we relax the definition [\(2.5\)](#) (cf. figure 7b).

References

1. Wootters WK, Zurek WH. 1982 A single quantum cannot be cloned. *Nature* 299, 802–803. ([doi:10.1038/299802a0](https://doi.org/10.1038/299802a0))
[Google Scholar](#) [Crossref](#) [Web of Science](#)
2. Marshall SM, Murray ARG, Cronin L. 2017 A probabilistic framework for identifying biosignatures using pathway complexity. *Phil. Trans. R. Soc. A* 375, 20160342. ([doi:10.1098/rsta.2016.0342](https://doi.org/10.1098/rsta.2016.0342))
[Google Scholar](#) [Crossref](#) [Web of Science](#)
3. Walker SI, Cronin L, Drew A, Domagal-Goldman S, Fisher T, Line M. 2019 Probabilistic biosignature frameworks. In *Planetary astrobiology* (eds Meadows V, Arney G, Schmidt B, Marais DJ). Tucson, AZ: University of Arizona Press. ([doi:10.2458/azu_uapress_9780816540068-ch018](https://doi.org/10.2458/azu_uapress_9780816540068-ch018))
[Google Scholar](#)
4. Meadows VS, Arney GN, Schmidt BE, Marais DJ. 2020 *Planetary astrobiology*. Tucson, AZ: University of Arizona Press; Lunar and Planetary



[Google Scholar](#)

5. Liu Y, Mathis C, Bajczyk MD, Marshall SM, Wilbraham L, Cronin L. 2021 Exploring and mapping chemical space with molecular assembly trees. *Sci. Adv.* 7, eabj2465. ([doi:10.1126/sciadv.abj2465](https://doi.org/10.1126/sciadv.abj2465))

[Google Scholar](#)

[Crossref](#)

[PubMed](#)

[Web of Science](#)

6. Marshall SM *et al.* 2021 Identifying molecules as biosignatures with assembly theory and mass spectrometry. *Nat. Commun.* 12, 3033. ([doi:10.1038/s41467-021-23258-x](https://doi.org/10.1038/s41467-021-23258-x))

[Google Scholar](#)

[Crossref](#)

[PubMed](#)

[Web of Science](#)

7. Marshall SM, Moore DG, Murray ARG, Walker SI, Cronin L. 2022 Formalising the pathways to life using assembly spaces. *Entropy* 24, 1099–4300. ([10.3390/e24070884](https://doi.org/10.3390/e24070884))

[Google Scholar](#)

[Crossref](#)

[PubMed](#)

[Web of Science](#)

8. Sharma A, Czégel D, Lachmann M, Kempes CP, Walker SI, Cronin L. 2023 Assembly theory explains and quantifies selection and evolution. *Nature* 622, 321–328. ([doi:10.1038/s41586-023-06600-9](https://doi.org/10.1038/s41586-023-06600-9))

[Google Scholar](#)

[Crossref](#)

[PubMed](#)

[Web of Science](#)

9. Jirasek Met *et al.* 2024 Investigating and quantifying molecular complexity using assembly theory and spectroscopy. *ACS Cent. Sci.* 10, 1054–1064. ([doi:10.1021/acscentsci.4c00120](https://doi.org/10.1021/acscentsci.4c00120))

[Google Scholar](#)

[Crossref](#)

[PubMed](#)

[Web of Science](#)

10. Raubitze S, Schatten A, König P, Marica E, Eresheim S, Mallinger K. 2024 Autocatalytic sets and assembly theory: a toy model perspective. *Entropy* 26, 808. ([doi:10.3390/e26090808](https://doi.org/10.3390/e26090808))

[Google Scholar](#)

[Crossref](#)

[PubMed](#)

[Web of Science](#)

11. Łukaszyk S, Bieniawski W. 2024 Assembly theory of binary messages. *Mathematics* 12, 1600. ([doi:10.3390/math12101600](https://doi.org/10.3390/math12101600))

[Google Scholar](#)

[Crossref](#)

[Web of Science](#)

12. Łukaszyk S. 2025 On the ‘Assembly Theory and its Relationship with Computational Complexity’. *IPIL Letters* 3, 1–6. ([doi:10.59973/ipil.157](https://doi.org/10.59973/ipil.157))

[Google Scholar](#)

13. Patarroyo KY, Sharma A, Seet I, Packmore I, Walker SI, Cronin L. 2025 Quantifying the complexity of materials with assembly theory. (<https://arxiv.org/abs/2502.09750>)

[Google Scholar](#)

14. Ziv J, Lempel A. 1978 Compression of individual sequences via variable-rate coding. *IEEE Trans. Inf. Theory* 24, 530–536. ([doi:10.1109/tit.1978.1055934](https://doi.org/10.1109/tit.1978.1055934))

[Google Scholar](#)

[Crossref](#)

[Web of Science](#)

15. Storer JA, Szymanski TG. 1982 Data compression via textual substitution. *J. ACM* 29, 928–951. ([doi:10.1145/322344.322346](https://doi.org/10.1145/322344.322346))
[Google Scholar](#) [Crossref](#) [Web of Science](#)
16. Welch. 1984 Technique for high-performance data compression. *Computer* 17, 8–19. ([doi:10.1109/MC.1984.1659158](https://doi.org/10.1109/MC.1984.1659158))
[Web of Science](#)
17. Charikar M, Lehman E, Liu D, Panigrahy R, Prabhakaran M, Sahai A, Shelat A. 2005 The smallest grammar problem. *IEEE Trans. Inf. Theory* 51, 2554–2576. ([doi:10.1109/TIT.2005.850116](https://doi.org/10.1109/TIT.2005.850116))
[Google Scholar](#) [Crossref](#) [Web of Science](#)
18. Kieffer JC, En-Hui Y. 2000 Grammar-based codes: a new class of universal lossless source codes. *IEEE Trans. Inf. Theory* 46, 737–754. ([doi:10.1109/18.841160](https://doi.org/10.1109/18.841160))
[Google Scholar](#) [Crossref](#) [Web of Science](#)
19. Kieffer J, En-hui Y, Park T, Yakowitz S. 1998 Complexity of preprocessor in MPM data compression system. In Proc. DCC '98 Data Compression Conf., Snowbird, UT, USA, 30 March–1 April 1998, p. 554. ([doi:10.1109/DCC.1998.672292](https://doi.org/10.1109/DCC.1998.672292))
20. Lehman E. 2002 Algorithms for grammar-based data compression. *PhD thesis*, Massachusetts Institute of Technology (MIT), Cambridge, MA, USA. <http://api.semanticscholar.org/CorpusID:11252484>.
[Google Scholar](#)
21. Nevill-Manning CG, Witten IH. 1997 Compression and explanation using hierarchical grammars. *Comput. J.* 40, 103–116. ([doi:10.1093/comjnl/40.2_and_3.103](https://doi.org/10.1093/comjnl/40.2_and_3.103))
[Google Scholar](#) [Crossref](#) [Web of Science](#)
22. Kieffer J, En-Hui Yang GN, Cosman P. 2000 Universal lossless compression via multilevel pattern matching. *IEEE Trans. Inf. Theory* 46, 1227–1245. ([doi:10.1109/18.850665](https://doi.org/10.1109/18.850665))
[Google Scholar](#) [Crossref](#) [Web of Science](#)
23. Kieffer J, Flajolet P. 2011 Universal lossless data compression via binary decision diagrams. (<https://arxiv.org/abs/1111.1432>)
[Google Scholar](#)
24. Nevill-Manning CG. 1997 Compression and explanation using hierarchical grammars. *Comput. J.* 40, 103–116. ([doi:10.1093/comjnl/40.2_and_3.103](https://doi.org/10.1093/comjnl/40.2_and_3.103))
[Google Scholar](#) [Crossref](#)
25. Larsson N. 1999 Offline dictionary-based compression. In Proc. DCC '99 Data Compression Conf., Snowbird, UT, USA, 29–31 March 1999, pp. 296–305. ([doi:10.1109/DCC.1999.755679](https://doi.org/10.1109/DCC.1999.755679))

26. Larsson NJ, Moffat A. 2000 Off-line dictionary-based compression. *Proc. IEEE* 88, 1722–1732. ([doi:10.1109/5.892708](https://doi.org/10.1109/5.892708))
[Google Scholar](#) [Crossref](#) [Web of Science](#)
27. Nevill-Manning CG. 1999 Compression and explanation using hierarchical grammars. *Comput. J.* 40, 103–116. ([doi:10.1093/comjnl/40.2_and_3.103](https://doi.org/10.1093/comjnl/40.2_and_3.103))
[Google Scholar](#) [Crossref](#)
28. Nevill-Manning CG, Witten IH. 1997 Identifying hierarchical structure in sequences: a linear-time algorithm. *J. Artif. Intell. Res.* 7, 67–82. ([doi:10.1613/jair.374](https://doi.org/10.1613/jair.374))
[Google Scholar](#) [Crossref](#) [Web of Science](#)
29. Apostolico A, Lonardi S. 2000 Off-line compression by greedy textual substitution. *Proc. IEEE* 88, 1733–1744. ([doi:10.1109/5.892709](https://doi.org/10.1109/5.892709))
[Google Scholar](#) [Crossref](#) [Web of Science](#)
30. Apostolico A, Lonardi S. 2000 Compression of biological sequences by greedy off-line textual substitution. In *Proc. DCC 2000: Data Compression Conf., Snowbird, UT, USA, 28–30 March 2000*, pp. 143–152. ([doi:10.1109/DCC.2000.838154](https://doi.org/10.1109/DCC.2000.838154))
31. Sakamoto H, Maruyama S, Kida T, Shimozone S. 2009 A space-saving approximation algorithm for grammar-based compression. *IEICE Trans. Inf. Syst.* E92-D, 158–165. ([doi:10.1587/transinf.E92.D.158](https://doi.org/10.1587/transinf.E92.D.158))
[Google Scholar](#) [Crossref](#) [Web of Science](#)
32. Takabatake Y, I T, Sakamoto H. 2017 Space-optimal grammar compression. *Leibniz Int. Proc. Inform.* 87, 1–67. ([doi:10.4230/LIPICS.ESA.2017.67](https://doi.org/10.4230/LIPICS.ESA.2017.67))
[Google Scholar](#)
33. Pagel S, Sharma A, Cronin L. 2024 Evolution of molecules across biochemistry with assembly theory. (<https://arxiv.org/abs/2409.05993>)
[Google Scholar](#)
34. Knuth DE. 2021 The art of computer programming. In *Seminumerical algorithms, vol. 2, 3rd edn* (ed. Donald EK). Boston, MA: Addison-Wesley.
[Google Scholar](#)
35. Clift NM. 2011 Calculating optimal addition chains. *Computing* 91, 265–284. ([doi:10.1007/s00607-010-0118-8](https://doi.org/10.1007/s00607-010-0118-8))
[Google Scholar](#) [Crossref](#) [Web of Science](#)
36. Cronin L. 2024 Exploring assembly index of strings is a good way to show why assembly & entropy are intrinsically different. See <https://x.com/leecronin/status/1850289225935257665>.
[Google Scholar](#)
37. Łukaszyk S. 2023 Black hole horizons as patternless binary messages and markers of dimensionality. In *Future relativity, gravitation, cosmology*, pp.

317–374. Hauppauge, NY: Nova Science Publishers. ([doi:10.52305/RLIT5885](https://doi.org/10.52305/RLIT5885))

[Google Scholar](#)

38. Łukaszyk S. 2024 Life as the explanation of the measurement problem. *J.*

Phys. Conf. Ser. 2701, 012124. ([doi:10.1088/1742-6596/2701/1/012124](https://doi.org/10.1088/1742-6596/2701/1/012124))

[Google Scholar](#)

[Crossref](#)

39. Łukaszyk S. 2025 Black hole merger as an event converting two qubits into

one. *Front. Quantum Sci. Technol.* 4, 1656200. ([doi:10.3389/frqst.2025.1656200](https://doi.org/10.3389/frqst.2025.1656200))

[Google Scholar](#)

[Crossref](#)

40. Gabric D, Shallit J, Zhong XF. 2021 Avoidance of split overlaps. *Discret.*

Math. 344, 112176. ([doi:10.1016/j.disc.2020.112176](https://doi.org/10.1016/j.disc.2020.112176))

[Google Scholar](#)

[Crossref](#)

[Web of Science](#)

41. Guibas LJ, Odlyzko AM. 1981 String overlaps, pattern matching, and

nontransitive games. *J. Comb. Theory Ser.* 30, 183–208. ([doi:10.1016/0097-3165](https://doi.org/10.1016/0097-3165)

(81)90005-4)

[Google Scholar](#)

[Crossref](#)

[Web of Science](#)

42. Ozelim L, Uthamacumaran A, Abrahão FS, Hernández-Orozco S, Kiani NA,

Tegnér J, Zenil H. 2025 Assembly Theory Reduced to Shannon Entropy and

Rendered Redundant by Naive Statistical Algorithms. *arXiv* 2408.15108. [https://ar](https://arxiv.org/abs/2408.15108)

[xiv.org/abs/2408.15108](https://arxiv.org/abs/2408.15108)

[Google Scholar](#)

43. Abrahão FS, Hernández-Orozco S, Kiani NA, Tegnér J, Zenil H. 2024

Assembly theory is an approximation to algorithmic complexity based on LZ

compression that does not explain selection or evolution. *PLoS Complex Syst.* 1,

e0000014. ([doi:10.1371/journal.pcsy.0000014](https://doi.org/10.1371/journal.pcsy.0000014))

[Google Scholar](#)

[Crossref](#)

44. Uthamacumaran A, Abrahão FS, Kiani NA, Zenil H. 2024 On the salient

limitations of the methods of assembly theory and their classification of

molecular biosignatures. *NPJ Syst. Biol. Appl.* 10, 82. ([doi:10.1038/s41540-024-00](https://doi.org/10.1038/s41540-024-00)

403-y)

[Google Scholar](#)

[Crossref](#)

[PubMed](#)

[Web of Science](#)

45. Vimal D, Parzych G, Smith OM, Parkar D, Bergen H, Daymude JJ, Mathis C.

2026 Open, reproducible calculation of assembly indices. *J. Open Source Softw.*

11, 9318. ([doi:10.21105/joss.09318](https://doi.org/10.21105/joss.09318))

[Google Scholar](#)

[Crossref](#)

46. Flamm C, Merkle D, Stadler PF. 2025 Assembly in directed hypergraphs.

Proc. R. Soc. A 481, 20250331. ([doi:10.1098/rspa.2025.0331](https://doi.org/10.1098/rspa.2025.0331))

[Google Scholar](#)

[Crossref](#)

[Web of Science](#)

47. Kempes CP, Lachmann M, Iannaccone A, Matthew Fricke G, Redwan

Chowdhury M, Walker SI, Cronin L. 2025 Assembly theory and its relationship

with computational complexity. *Npj Complex.* 2, 27. ([doi:10.1038/s44260-025-00](https://doi.org/10.1038/s44260-025-00)



049-9)

[Google Scholar](#)

[Crossref](#)

[PubMed](#)

48. Masierak P. 2026 Computational complexity of determining the assembly index. *IPI Letters* 4. ([10.59973/ipil.315](#))

[Google Scholar](#)

49. Gebhard TDet *al.* 2022 Inferring molecular complexity from mass spectrometry data using machine learning. In Machine Learning and the Physical Sciences Workshop, NeurIPS 2022, New Orleans, LA, USA. NeurIPS Foundation.

50. Cronin L, Parra JCM, Patarroyo KY. 2025 Assembly addition chains. (<http://arxiv.org/abs/2512.18030>)

[Google Scholar](#)

51. Krzyżanowski W. 2025 Procesy ewolucji kulturowej muzyki w środowisku technologii cyfrowych. Uniwersytet Adama Mickiewicza, Poznań, Poland. ([doi:10.31237/osf.io/wkc4z_v1](#))

[Google Scholar](#)

52. Vopson MM. 2023 The second law of infodynamics and its implications for the simulated universe hypothesis. *AIP Adv.* 13, 105308. ([doi:10.1063/5.0173278](#))

[Google Scholar](#)

[Crossref](#)

[Web of Science](#)

53. Grassberger P. 2012 Randomness, information, and complexity. (<https://arxiv.org/abs/1208.3459>)

[Google Scholar](#)

54. Bieniawski W, Masierak P, Tworz S, Tomski A, Łukaszyk S. 2026 *Assembly theory—formalizing assembly spaces, discovering patterns and bounds*. GitHub. See <https://github.com/szluk/AssemblyTheory>.

55. Bieniawski W, Masierak P, Tworz S, Tomski A, Łukaszyk S. 2026 szluk/Assemblytheory: initial release v1_(30.03.2026). Zenodo. ([doi:10.5281/zenodo.19331598](#))

[Google Scholar](#)

56. Bieniawski W, Masierak P, Tomski A, Łukaszyk S, Tworz S. 2026 Supplementary material from: Assembly Theory - Formalizing Assembly Spaces and Discovering Patterns and Bounds. Figshare. ([doi:10.6084/m9.figshare.c.8490332](#))

[Google Scholar](#)

© 2026 The Authors.

Published by the Royal Society under the terms of the Creative Commons Attribution License <http://creativecommons.org/licenses/by/4.0/>, which permits unrestricted use, provided the original author and source are credited.



Supplementary data

Supplementary Material from Assembly Theory - Formalizing Assembly Spaces and Discovering Patterns and Bounds

42

0

10

views

shares

downloads



 [figshare](#) _

1 / 13

sp1- zip file



0 Comments

 Login ▾

G

Start the discussion...

LOG IN WITH

OR SIGN UP WITH DISQUS 

Name

 3

Share

Best Newest Oldest

Be the first to comment.

[Subscribe](#) [Privacy](#) [Do Not Sell My Data](#)

